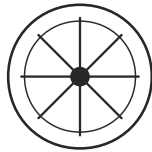


THE CURIOUS FAMILY LIBRARY

How Cars Work

A book for curious passengers



MEIRA DAVIDSON

FOR THE JESSICA FAMILY KIDS

be curious, stay open, be kind.

They are the reason I started making books for kids.

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First edition.

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PROLOGUE

Opening the Box

Look at a toy car for a minute. The kind that fits in one hand. You can see the wheels turning when you push it across the floor. You can flip it over and see how the wheels are mounted on a thin metal axle. Push it harder and it goes farther; push it on carpet and it barely goes at all. Nothing is hidden. Toy cars make sense.

Now think about a real car.

You climb into the back seat. The door slams. Your mom turns a key or pushes a button. The car makes a sound — maybe a quiet hum, maybe a deeper rumble — and then it moves. It carries you and your mom and your backpack and a week of groceries down the street. It does this for an hour, or for fifty miles, or for as long as anyone wants. Nobody is pushing it. There is no obvious reason it should be moving.

Real cars do not make sense the way toy cars do. They are sealed boxes. The wheels turn, and the car goes, but you cannot see why.

This book is about why.

We are going to open the box. Not literally — your mom will not be thrilled if you start unscrewing the hood — but in your head. By the end of the book, when you hear a car start, you will know what is actually happening inside it. When the brake lights come on at a stop sign, you will know what is grabbing the wheels. When you pass an electric car at a charging station, you will know what is flowing into its battery and what that battery is actually doing.

Three parts

The first part is about old cars — cars from your grandparents' time, the kind made of metal and rubber and not much else. These cars are mostly mechanical. That means almost everything that happens, you could see happening if you took the car apart. Pieces push other pieces. Gears turn other gears. There is a logic to it that you can follow with your eyes.

The second part is about how cars got smart. Sometime around the 1980s, people started putting computers in cars. At first just little ones, doing small jobs. Now there are dozens of computers in a regular car, talking to each other, watching the road, deciding things in fractions of a second. Modern cars are still mechanical underneath, but the mechanical parts are managed by software.

The third part is about electric cars. These are different. No engine, no gasoline, no exhaust pipe. A giant battery and an electric motor and not much else. Simpler in some ways, stranger in others.

How to read it

Each chapter has a Try It — something you can actually do, usually with stuff already in the kitchen. A candle and a jar. A balloon. A magnet. A bike. These are not extras. They are the point. Reading about how something works will only get you so far. Doing the small version yourself is what makes it stick.

There are also blue boxes scattered through the chapters with surprising stories or odd facts about the people and machines involved. Read them or skip them. They are not on the test, because there is no test.

You don't drive yet. That's fine. You don't need to drive to understand a car, the same way you don't need to fly a plane to understand how a wing works. You just need to pay attention.

Let's open the box.

PART ONE

How an Old Car Works

Suck, Squeeze, Bang, Blow

A car is heavy. Heavier than you, heavier than your mom, heavier than your mom and you and a full week of groceries combined. A normal car weighs about three thousand pounds. That is roughly the same as a small horse, except a horse can move itself and a car cannot, not on its own.

Something has to push it.

When you ride a bicycle, you know what is pushing it: you are. Your legs push the pedals, the pedals turn the chain, the chain turns the back wheel, and the bike rolls forward. If you stop pushing, the bike slows down and stops. The connection between your legs and the wheels is direct. You can feel it.

A car has no pedals you push. (It has pedals your mom pushes with her foot, but those don't push the wheels — they just tell the car what to do. We will get to those later.) So what is pushing a car?

The answer is small explosions. Hundreds of them, every second. Every car like the one your grandmother drove, and most cars on the road today, runs on a controlled, repeating series of tiny explosions inside the engine. The explosions push metal pieces. The metal pieces turn other metal pieces. Eventually the wheels turn, and the car moves.

That is the whole story. The rest of this chapter is the details.

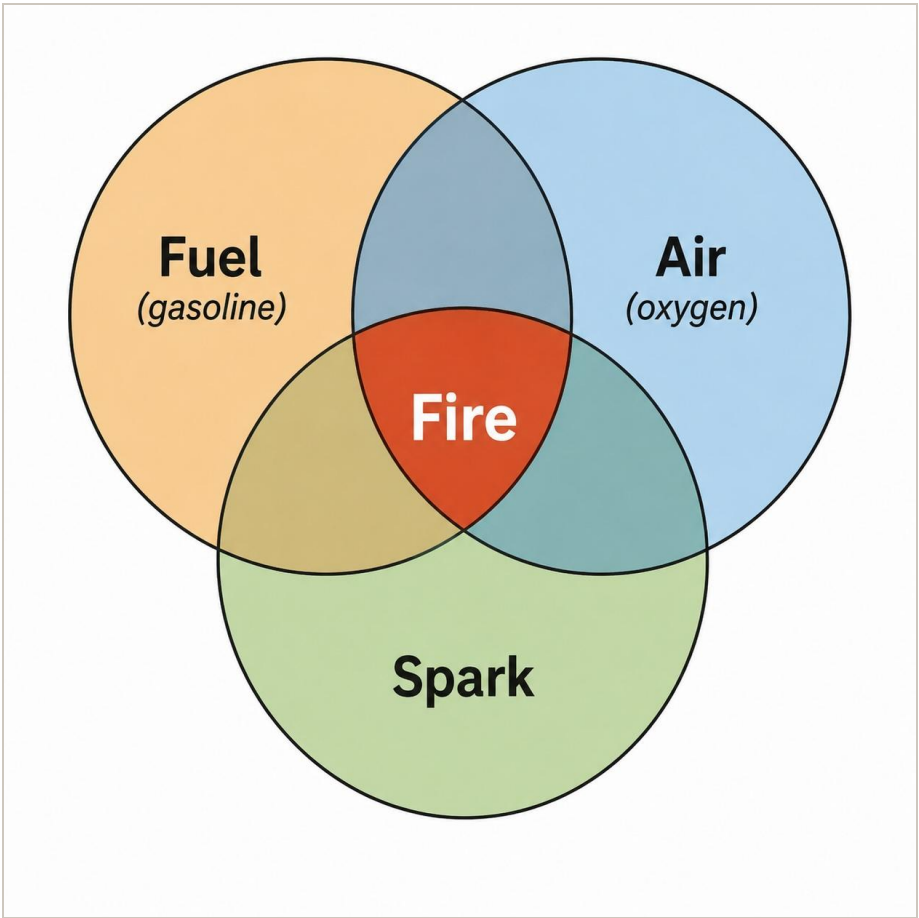
Fire needs three things

To have an explosion, you need fire. To have fire, you need three things: something to burn, something to burn it with, and something to start it.

Something to burn is gasoline. Gasoline is a liquid made from oil that comes out of the ground, and it burns very, very well. If you have ever stood near a pump at a gas station, you have smelled it.

Something to burn it with is air. Specifically, the oxygen in air. About one-fifth of the air around you is oxygen, and oxygen is what makes things burn. Without oxygen, fire cannot exist.

Something to start it is a spark. Gasoline and air sitting together don't do anything. Mix them in a jar and ignore them and they will just sit there. You have to give them a kick. A small electric spark is enough.



Fire needs three things: fuel, air, and a spark.

TRY IT: THE CANDLE AND THE JAR

You will need:

- A small candle (a tea light works perfectly)
- A glass jar that is bigger than the candle (a clean pasta-sauce jar is great)
- A grown-up to help with the lighting

Put the candle on a plate. Light it with the grown-up's help and let it burn for a few seconds. Then turn the jar upside down and lower it over the candle, so the mouth of the jar sits flat on the plate and the candle is sealed inside.

Watch what happens. The flame keeps burning for a few seconds. Then it gets smaller. Then it gets very small. Then it goes out. There is still candle left. There is nothing wrong with the wick. The candle did not run out of fuel.

What ran out is the oxygen. Once the flame burned through all the oxygen sealed inside the jar, the fire could not continue. Take the jar off, light the candle again, and it burns happily, because now the air can get to it.

This is exactly what is happening inside a car engine, except instead of a candle you have gasoline, and instead of an open flame you have a controlled explosion. No oxygen, no boom. The whole engine is built around making sure the right amount of fuel meets the right amount of air at exactly the right time.

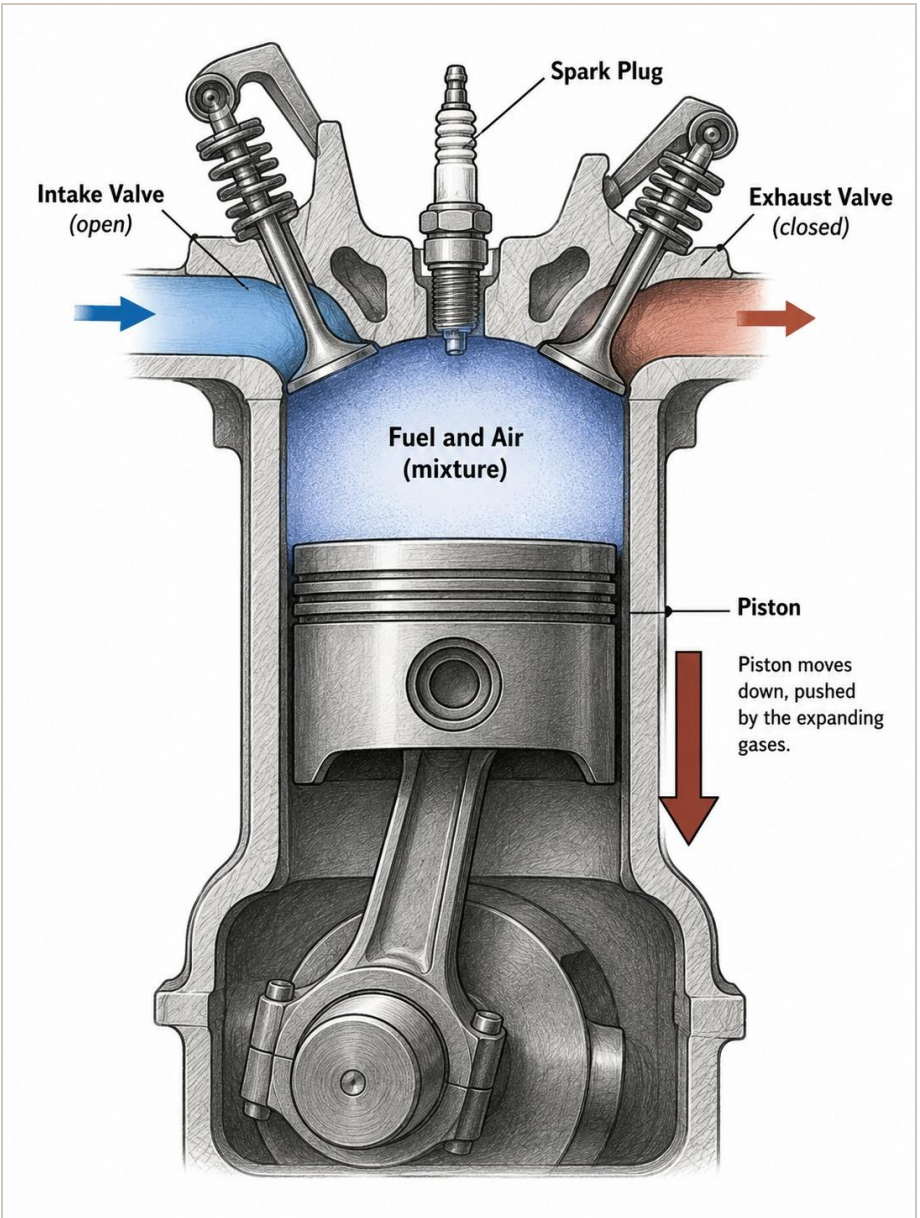
A metal cup in a metal tube

Now we need something for the explosion to push.

Picture a metal cup that fits snugly inside a metal tube — close enough that nothing leaks past the edges, but loose enough that the cup can slide up and down freely. The cup is called a **piston**. The tube is called a **cylinder**.

If you light a small amount of gasoline-and-air mixture in the closed top of the cylinder, the explosion shoves the piston down. Hard. Faster than you can blink.

That is the only thing the explosion has to do. Push the piston down. Everything else in the engine is just a clever way to set up the next explosion and turn that downward push into something useful.



A single cylinder during the BANG stroke.

The four strokes

You can't have one explosion and call it done. Cars need to keep going for hours. So the engine has to repeat this same trick — explosion, push, explosion, push — over and over.

To do that, every cylinder repeats the same four-step cycle, called the **four-stroke cycle**. Each stroke is one movement of the piston, either up or down. The four strokes have nicknames, and the nicknames are accurate and a little funny:

1. SUCK. The piston moves down. A little door at the top of the cylinder (called an **intake valve**) opens, and the falling piston pulls in a fresh mixture of air and gasoline. Like pulling on the plunger of a syringe.

2. SQUEEZE. The little door slams shut. The piston moves back up, squishing the air-and-gasoline mixture into a tiny space at the top. Squeezing a gas makes it hot and ready to burn.

3. BANG. A **spark plug** — a small device that can shoot an electric spark on demand — fires. The squished mixture explodes. The piston gets shoved down hard. This is the only stroke where the engine is making power. The other three are just setting it up.

4. BLOW. The piston comes back up. A different little door (the **exhaust valve**) opens, and the piston pushes the burned, used-up gas out of the cylinder. Then that door slams shut, the intake door opens again, and we are back at SUCK.

Down, up, down, up. Suck, squeeze, bang, blow. Two full trips up and back down for the piston, and the engine has produced one push.

A typical car engine does this somewhere between ten and a hundred times every second, in each cylinder. When you hear a car running, what you are hearing is hundreds of these little cycles per second, blurred into one steady sound.

DID YOU KNOW? The four-stroke cycle was invented by a German engineer named Nicolaus Otto in 1876. Engineers still call it the Otto cycle in his honor, almost 150 years later. Otto wasn't even a trained engineer — he started out as a grocer who got obsessed with engines in his spare time and eventually built one that worked better than anyone else's. Almost every gasoline car ever made uses a version of his idea.

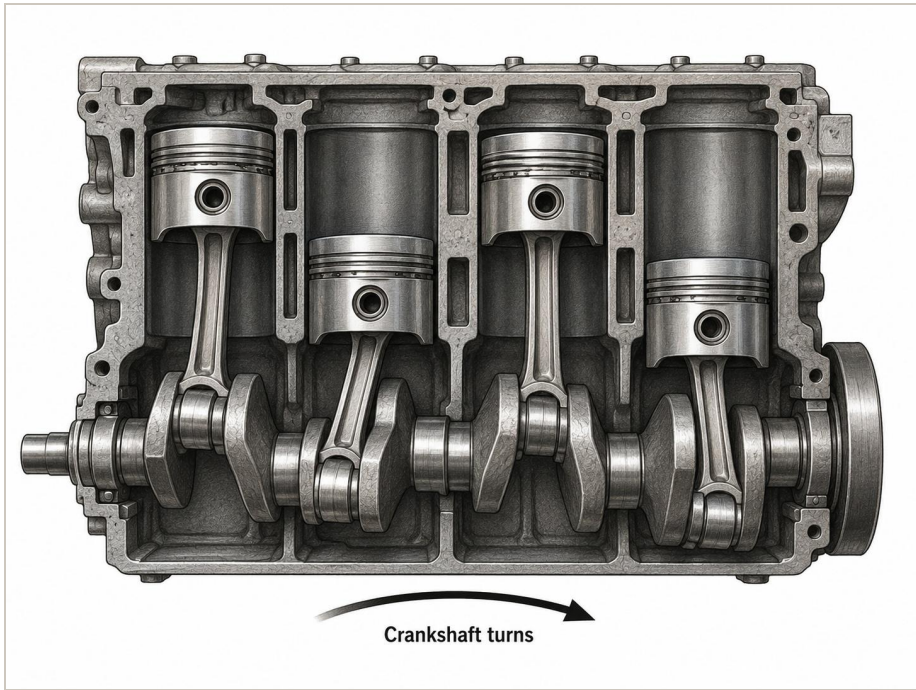
Four cylinders, taking turns

A regular car doesn't have one cylinder. It has four, or six, or eight, all running this same dance, but timed so they take turns. While cylinder 1 is on BANG, cylinder 2 might be on SQUEEZE, cylinder 3 on BLOW, and cylinder 4 on SUCK. They are staggered so that there is always at least one cylinder making power at any given moment.

Think of a Lego model where four little arms all push down on the same gear, but never at the same time. Each arm does its job in turn, and the gear keeps spinning smoothly. That is an engine.

All the pistons push down on the same long metal rod called a **crankshaft**. The crankshaft has the shape of a few zigzags — bent in the right places so that when a piston pushes straight down, the crankshaft has to turn sideways to make room. Up-and-down motion at the top becomes round-and-round motion at the bottom.

Just like your legs on a bicycle. Your knees go up and down, but the pedals turn in circles. The connecting bar between your knee and the pedal makes the conversion. The crankshaft does the same thing in a car, except with metal arms and explosions instead of your legs and a peanut butter sandwich.



Four pistons and a crankshaft, taking turns.

That's the heartbeat

When you hear a car start, what you are hearing is the four-stroke cycle beginning. A small motor (called the starter) turns the engine over a few times to get the cycle going, and then the explosions take over and the engine runs itself, banging along at thousands of cycles a minute.

When the car is sitting at a red light and you can hear it idling — that low, even rumble under the hood — that is the four-stroke running slowly, just enough to keep itself going. When your mom presses the gas pedal, more fuel and more air flow in, the bangs come faster and harder, and the wheels turn faster.

The whole rest of an old-fashioned car is built around this single trick: the heartbeat in the middle, where small, controlled explosions

happen thousands of times a minute, and somehow — except it isn't somehow, it is the next four chapters — that heartbeat ends up turning the wheels under your seat.

The Mix

Last chapter, we said the engine sucks in a mixture of air and gasoline, lights it on fire, and uses the explosion to push the piston down. That sen-

tence skipped a real question. Where does the mixture come from?

There is no can of pre-mixed fuel-and-air sitting in the trunk. Cars don't come with one. The mixture has to be made fresh — in the right amounts, hundreds of times every second, on its own — and dropped into each cylinder right when the cylinder needs it.

On an old car, the device that does this is called a **carburetor**. (Pronounced KAR-bur-ay-ter. Some people drop the middle and just say “carb.”) Before we get to it, we should know how the gasoline reaches it in the first place.

From the tank to the engine

Gasoline lives in a metal tank, usually mounted under the back of the car. When you go to a gas station, that's what's getting filled up. A typical tank holds twelve to twenty gallons — enough fuel for a few hundred miles.

But the engine is at the front of the car, ten or fifteen feet from the tank. The gasoline has to travel that distance. It does this through a thin metal pipe called a **fuel line**, which runs along the bottom of the car like a vein.

Pushing the fuel through the pipe is a **fuel pump**. On an old car, the fuel pump is a small mechanical device bolted to the engine. It

works by being squeezed and released by a part of the engine that is already moving anyway. Every time the engine moves, the pump pumps a little gasoline forward toward the front. When the engine stops, the pump stops. Simple.

Modern cars use electric fuel pumps that can run even when the engine is off, but for our 1960s car, no battery, no electricity — just the engine pulling its own fuel forward.

Why mixing matters

Liquid gasoline, by itself, does not explode. A puddle of gasoline lit on fire just burns slowly, with a flame on the surface. There is no boom. To get the kind of fast, hard explosion that can push a piston down, the gasoline has to be in the form of tiny droplets suspended in air, like a mist. Think of bug spray, or perfume coming out of an old-fashioned perfume bottle, or fog from a fog machine. That is the form gasoline has to be in to burn explosively.

It also has to be mixed with air in the right proportion. Too much fuel and not enough air, and the mixture gets sooty and won't ignite cleanly. Too much air and not enough fuel, and there's nothing there to burn. The sweet spot is roughly fourteen parts of air to one part of fuel by weight. For every spoonful of gasoline, about fourteen spoonfuls of air. Off by much in either direction and the engine starts running badly or stops altogether.

So the carburetor has two jobs: turn liquid gasoline into mist, and mix that mist with air in the right ratio. Both at the same time, automatically, every cycle, on its own.

Fast air, low pressure

Here is the trick the carburetor uses, and it is one of the prettiest pieces of physics in any car.

When air moves fast, its pressure drops.

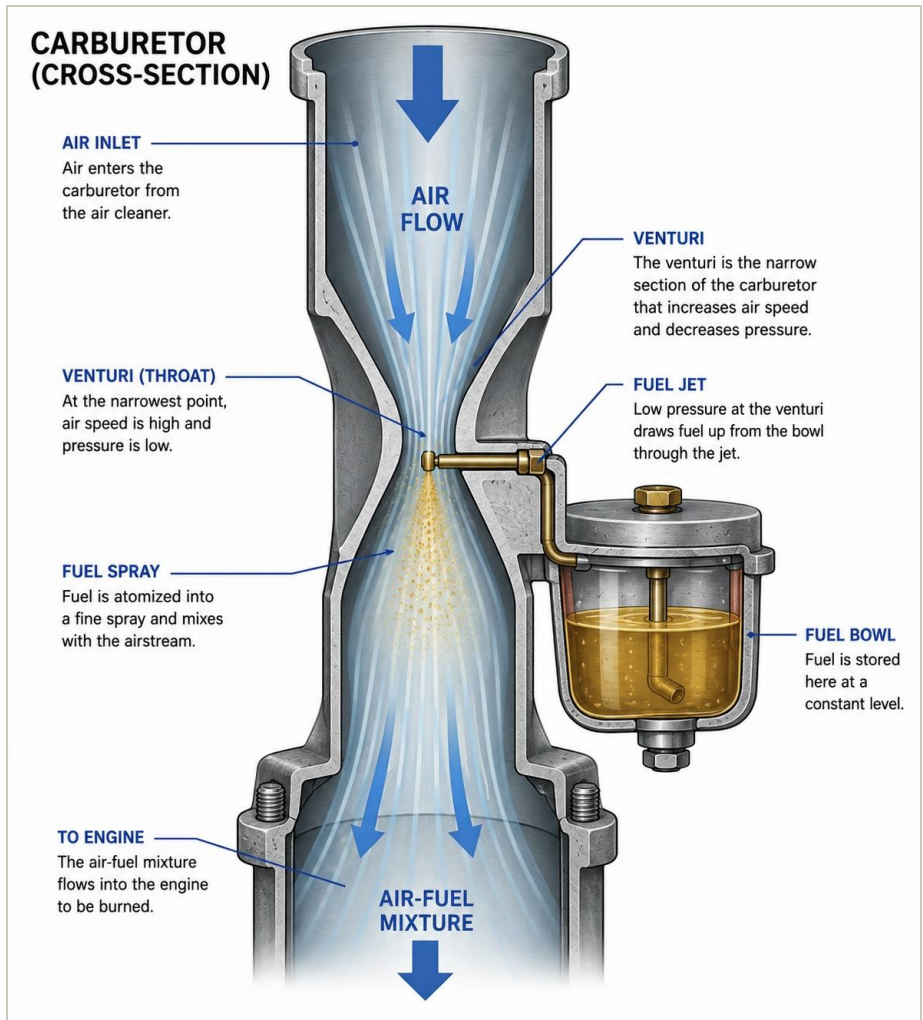
That sentence sounds like nonsense the first time you read it. It is true anyway. Air molecules in slow-moving air have time to push against everything around them — including outward, sideways, on the walls of whatever they're flowing through. Air molecules in fast-moving air are too busy moving forward to do much pushing sideways. Less sideways push means lower pressure.

The carburetor takes advantage of this. It is a tube with a narrowed section in the middle, like an hourglass laid on its side. Air gets sucked through the tube by the engine. (Remember the SUCK stroke from last chapter? That is what is doing the sucking.) When air rushes through the narrow part, it has to speed up to fit through the smaller space.

Right at the narrowest point, the engineers have drilled a tiny hole. The hole connects, by a small passage, to a little reservoir of gasoline fed by the fuel pump.

Here is what happens. The air rushing through the narrow throat is moving fast, so its pressure is low. The gasoline in the reservoir is at normal air pressure, which is higher than low pressure. Higher pressure pushes toward lower pressure, so the gasoline gets shoved up through the tiny hole and squirts into the fast-moving airstream. But it doesn't just dribble out. The fast air rips it into a fine mist on the way past.

By the time the air leaves the carburetor, it is full of perfectly suspended droplets of gasoline, ready to burn. That is the entire trick. Fast air, low pressure, fuel sucked up, atomized, mixed.



A carburetor, in cross-section.

TRY IT: THE STRAW ATOMIZER

You will need:

- A glass of water
- Two regular drinking straws

Stick one straw straight down into the glass of water, with the top of it sticking up about an inch above the water surface.

Take the second straw and hold it horizontally, with one end right next to the top of the first straw — almost touching, but at a right angle, making a T-shape.

Now blow hard through the horizontal straw, across the top of the vertical one. Don't blow into it — blow across it.

Water rises up the vertical straw and sprays out the top as a mist. Sometimes it shoots out hard enough to soak whatever is in front of you. (Maybe do this over a sink.)

You just made a carburetor. The fast air from your breath, blowing across the top of the vertical straw, lowered the pressure right above it. The water below was at normal air pressure. Higher pressure pushed the water up. The fast air ripped it into a mist on its way past.

That is exactly what is happening inside a carburetor on a running engine, except the air is being sucked through by the engine instead of blown by your lungs, and the liquid is gasoline instead of water. Same physics. Same result.

What the gas pedal actually does

There is one more part of this story, and it explains what your mom is really doing when she presses the gas pedal.

She is not pumping fuel into the engine. The carburetor takes care of that on its own.

She is opening a flap.

Inside the carburetor, downstream of the narrow throat, there is a flat metal disc on a hinge. When the disc is sideways across the tube, almost no air can get past, and the engine runs slowly — barely getting any air, barely making any power, just enough to keep itself going. (This is what the engine is doing when the car is sitting at a red light.) When the disc is tilted edge-on to the airflow, the way is wide open,

air rushes through, and the engine runs fast. The disc is called the **throttle**.

The gas pedal is connected by a metal cable to the throttle. Press the pedal, the cable pulls, the throttle swings open, more air rushes through the carburetor, and — this is the clever part — more fuel automatically comes with it. Because the carburetor uses fast air to suck up fuel, more air means more fuel, in roughly the right ratio. The mix takes care of itself. Press harder, more air, more fuel, more bang, faster wheels.

In other words, the gas pedal does not really control the gas. It controls the air. The gas just tags along.

This is also why an old car's gas pedal is sometimes called the throttle pedal. That is what it actually is.

DID YOU KNOW? Until the 1980s, every car had a small knob on the dashboard called the choke. On a cold morning, before turning the key, a driver would pull the choke knob out, then push it back in slowly as the engine warmed up. The choke was a second flap inside the carburetor that, when closed, blocked some of the incoming air. Less air through the throat made the mixture richer (more fuel, less air), which a cold engine needed in order to start. As the engine warmed up, the driver would gradually open the choke until the engine was running on a normal mixture again. Modern cars handle all this automatically with computers and sensors. But for sixty years, “choking the engine” in cold weather was just part of starting a car.

Ready to burn

Now we know what is happening inside the cylinder during the SUCK stroke. The piston pulls down. The throttle is open by however much your mom is pressing the pedal. Air rushes through the carburetor. Fuel gets pulled into the airstream and torn into a mist. The

cylinder fills up with a precise haze of gasoline-and-air vapor, ready to be squeezed and exploded.

The next thing the engine needs is the spark.

That is Chapter 3.

The Spark

We left the cylinder full of mist. The piston has just finished the SQUEEZE stroke — the gasoline-and-air mixture is compressed into a tiny space at the top of the cylinder, hot from the squeezing, ready to burn.

It still needs a kick. Without that kick, the mixture will just sit there, hot but not burning, and the engine won't run. From last chapter you already know what the kick is: a spark. A small electric spark, jumping across a gap inside the cylinder, at exactly the right millisecond.

In a fast-running engine, this happens in each cylinder somewhere between thirty and a hundred times per second. Every spark has to land in the right cylinder at the right phase of the four-stroke cycle. If a spark fires too early, the mixture explodes before the piston has finished squeezing. If it fires too late, the piston has already started moving back down. Either way, the engine runs badly or stops.

This whole chapter is about how an old car pulls off this trick.

What a spark actually is

If you have ever shuffled across a carpet in socks and then touched a metal doorknob, you have made a spark. You felt the zap. You may even have seen the flash, in a dark hallway. That is a spark. Electricity, briefly jumping through air.

This is supposed to be impossible. Normally, air does not conduct electricity. Engineers call air an insulator — a substance that doesn't

let current flow through it. That is why bare power lines in the sky don't continuously zap each other. The air between them is in the way.

But under enough pressure — the right kind of pressure — air gives up. If you push electricity hard enough across a small gap of air, the air will briefly become a conductor and let the current through. That moment of giving up is the spark you see and feel.

The “pressure” in this case is voltage. Voltage is how hard the electricity is being pushed. A typical AA battery in a TV remote pushes at one and a half volts. A wall outlet pushes at one hundred and twenty volts. A lightning bolt is somewhere around a hundred million volts.

The amount of voltage it takes to make a spark depends on the size of the gap. For the tiny gap inside a spark plug — about the width of a paperclip wire — it takes around twenty thousand volts.

Twenty thousand volts. The car has a battery that makes twelve. We are going to need to do some work to get from twelve to twenty thousand.

How a coil makes more

The work is done by a device called the **ignition coil**. From the outside, it looks like a small black can about the size of a soda can. Inside, it is two long pieces of wire, each wound into a tight coil, both wrapped around the same iron rod.

The two coils have very different lengths. One — called the primary coil — has maybe two hundred wraps of fairly thick wire. The other — the secondary coil — has tens of thousands of wraps of much thinner wire, wound around the outside of the primary.

Here is how it works. When current from the battery flows through the primary coil, the iron rod becomes a magnet. (Coils of wire with current flowing through them act like magnets. Wrap a wire around a nail, hook the ends of the wire to a battery, and the nail will pick up

paper clips. Same idea.) The magnetic field threads through the inner coil and also through the much bigger outer coil.

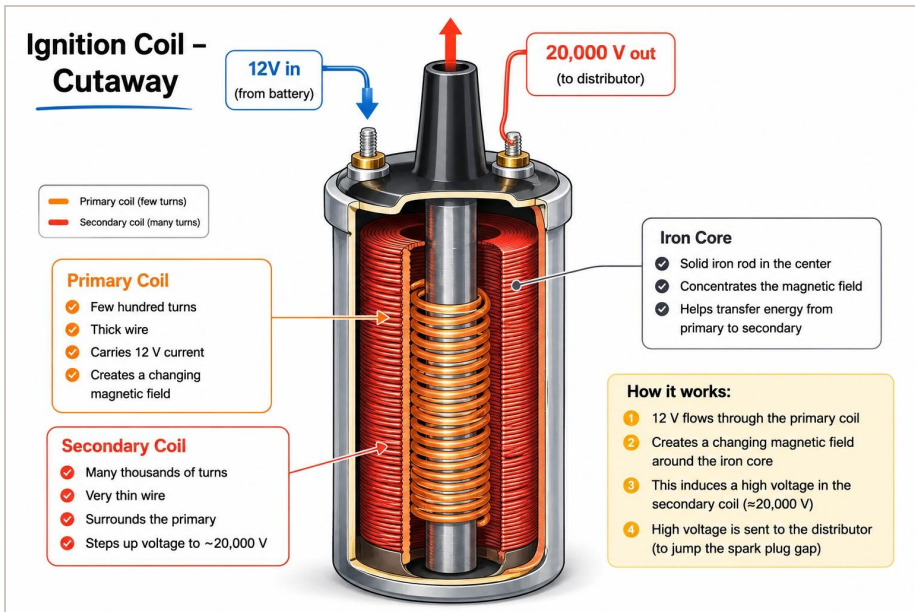
Now the clever part. As long as the current is flowing steadily, nothing interesting happens with the second coil. But if you suddenly turn the current off — if you cut it abruptly — the magnetic field collapses. And here is the rule of physics that makes everything work:

A collapsing magnetic field through a coil of wire creates a voltage in that coil. The faster the field collapses, and the more turns of wire the coil has, the bigger the voltage.

The secondary coil has tens of thousands of turns of wire. When the magnetic field built up by the primary suddenly collapses, the secondary's many turns each get a tiny voltage, and they all add up to one giant voltage at the ends of the wire.

The math works out about the way you would expect. If the secondary has a hundred times as many turns as the primary, the voltage at the secondary's ends will be about a hundred times the voltage that was driving the primary. Twelve volts in, twelve hundred volts out. Or twelve volts in and twelve thousand volts out, if the ratio is bigger.

In a real ignition coil, the ratio is about a thousand to one. Twelve volts of battery current goes in. When it gets cut off, twenty thousand volts springs out the other end.



Inside an ignition coil.

Cutting the current

For all this to work, the current through the primary coil has to be cut off — fast — once per spark. In a 1960s car, this is done by a small mechanical switch called the **points**. Two metal contacts that touch each other most of the time, but get pulled apart by a small bump on a rotating shaft inside the engine. Touch, current flows, magnetic field builds. Pull apart, current stops, field collapses, twenty thousand volts springs out of the secondary coil.

The shaft that opens and closes the points is driven by the engine itself. Every time the engine rotates, the points click open at exactly the moment one of the cylinders needs a spark. The whole system stays in sync because all of it — the points, the engine, the pistons — is connected to the same rotating parts.

Modern cars use electronic switches with no moving parts. But for sixty years, every spark in every car came from two pieces of metal

touching and untouching, dozens of times per second, on a moving rod inside the engine.

Sending it to the right cylinder

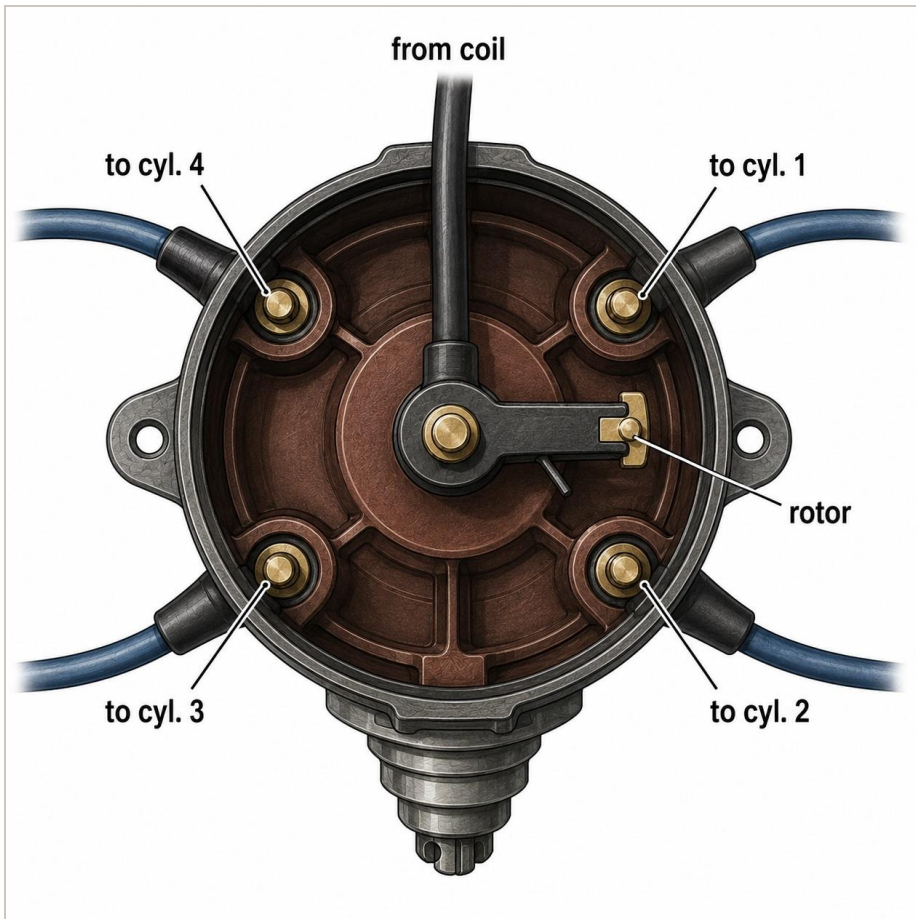
There is still a problem. The ignition coil makes one giant pulse of high voltage every time the points open. But the engine has multiple cylinders, each needing a spark in turn. Cylinder 1 needs a spark, then cylinder 3, then cylinder 4, then cylinder 2 (or whatever order — it varies by engine). The pulse from the coil has to be sent to the right spark plug at the right time.

This is the job of the **distributor**.

A distributor is a small cylinder that sits on top of the engine, with a rotating arm inside. The arm is called the **rotor**, and it spins at half the speed of the engine. (Half, because each cylinder fires once every two engine rotations — once per four-stroke cycle.) Around the inside of the cap that covers the rotor, there are little metal contacts — one for each spark plug — and a thick wire runs from each contact to the matching spark plug down in its cylinder.

In the center of the cap, a wire from the ignition coil delivers the twenty-thousand-volt pulse. The pulse comes down the center, jumps a tiny gap onto the rotor, and the rotor — wherever it happens to be pointing at that instant — passes the pulse on to the contact directly under its tip. From there, it shoots down the wire to the right spark plug.

No computer. Just a spinning metal arm, sending electricity to whichever cylinder it happens to be pointing at.



A distributor cap, viewed from above.

Where the spark actually happens

At the end of the chain is the spark plug. We met it briefly in Chapter 1; here is what is actually inside it.

A spark plug is a small ceramic-and-metal device, about the size and shape of a thick crayon, that screws into the top of each cylinder. The bottom end pokes down into the cylinder. The top end sticks out where the wire from the distributor can connect to it. Inside the spark plug, the wire runs down through a ceramic insulator and ends in a

small metal post called the **electrode**. Right next to the electrode, hooked to the body of the plug (which is in turn screwed into the metal of the engine), is a second metal post. There is a tiny gap between them, about the width of a paperclip wire.

When the twenty-thousand-volt pulse arrives, it jumps across that gap, through the gasoline-and-air mist, and ignites it.

That is the spark.

TRY IT: THE BALLOON AND THE DOORKNOB

You will need:

- A latex balloon, blown up and tied
- A clean head of hair (yours)
- A metal doorknob
- A dark or dim room

Rub the balloon vigorously back and forth on top of your head for ten or fifteen seconds. Your hair will probably try to follow the balloon when you pull it away. That is a clue. You have moved electrons from your hair onto the balloon. The balloon is now slightly negatively charged. Your hair is slightly positively charged.

In a dark room, slowly bring the charged balloon close to a metal doorknob — but not quite touching. Watch the gap.

If you get close enough, you will see a tiny spark and hear a faint snap. The electrons crowded onto the balloon, pushed by the voltage of being concentrated all in one place, found a path through the air to the metal of the doorknob. Air briefly stopped insulating and started conducting.

That is the same kind of event that happens between the two electrodes of a spark plug, scaled differently. The spark plug uses an ignition coil to build the voltage. You used your hair. The physics is

identical: enough voltage, narrow enough gap, air gives up, lightning happens.

DID YOU KNOW? A lightning bolt is the same kind of thing that happens between the two electrodes of a spark plug. Charge builds up — in lightning's case, between a cloud and the ground, or between two clouds — until the voltage is so high that air briefly conducts electricity. The spark from a spark plug is about a quarter of an inch long. A lightning bolt is several miles long. The voltage scales accordingly: about twenty thousand volts for the spark plug, around a hundred million volts for the lightning. Same physics, very different size. Both look about as bright if you stare directly at them, but please don't stare directly at lightning.

The whole chain, in one breath

The battery sends twelve volts through the points and into the primary coil. A magnetic field builds up around the iron rod in the middle. At the right instant in each cylinder's cycle, the engine pushes the points apart, the current stops, the magnetic field collapses, and twenty thousand volts springs out the other end of the secondary. That pulse travels down the center wire of the distributor, jumps to the rotor, and is passed to whichever spark plug the rotor happens to be pointing at. The pulse runs down the spark plug wire, into the spark plug, jumps the tiny gap inside the cylinder, and ignites the mist.

BANG.

Two strokes left in the cycle. The piston is now flying down toward the bottom of the cylinder, and somehow this turns into the wheels rolling under your seat. How? That is Chapter 4.

From Bang to Wheels

BANG. The piston flies down toward the bottom of the cylinder, riding the explosion. Through the connecting rod, it pushes the crankshaft around, the way your knee pushes a bicycle pedal.

Now we have a long, heavy metal rod, spinning fast. The wheels are over there. We need to get the spinning of the rod to the spinning of the wheels.

This sounds easy. It is not.

The problem

A car engine is fussy. It works best when its crankshaft is spinning at a particular range of speeds — usually between 1,000 and 5,000 turns per minute, which mechanics shorten to RPM. Below that range, the engine doesn't have enough momentum to keep itself going and stalls out. Above it, it starts shaking itself apart.

The wheels of the car, on the other hand, need to spin at very different speeds depending on what the car is doing. When the car is parked, the wheels are at zero RPM. Creeping out of a parking spot, maybe 50 RPM. On the highway at 65 miles per hour, the wheels are spinning at about 800 RPM.

So we have an engine that wants to spin between 1,000 and 5,000 RPM, and wheels that need to spin between 0 and 800 RPM. These ranges don't match. They don't match at any speed.

What do we do?

Gears

The answer is gears. A gear is a wheel with teeth around its edge. When two gears are meshed together — their teeth interlocked — turning one of them turns the other.

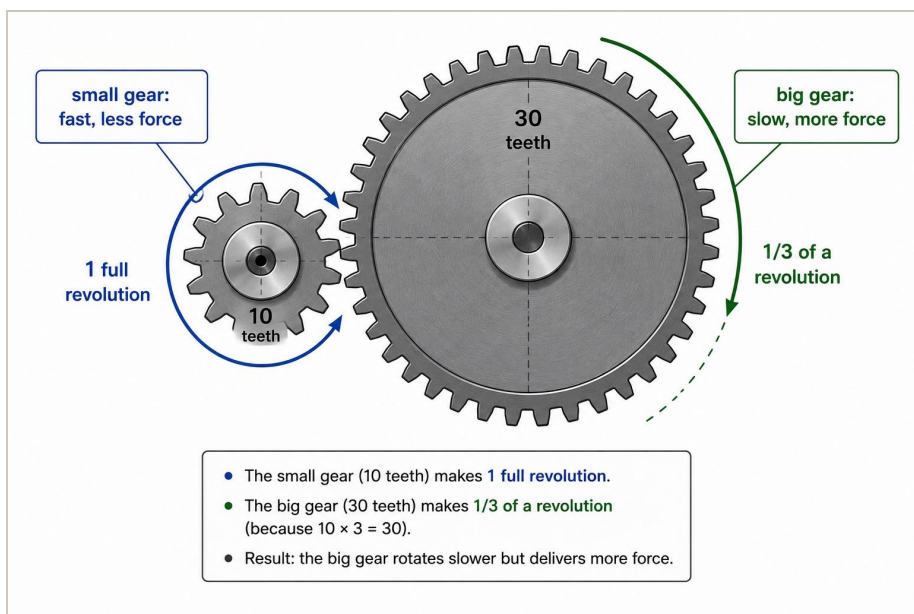
The thing that makes gears interesting is what happens when the two meshed gears are different sizes.

Picture a small gear with ten teeth, meshed against a big gear with thirty teeth. If you turn the small gear all the way around once, you push exactly ten teeth past the meeting point. The big gear has to turn far enough to receive those ten teeth on its own side — which, since it has thirty teeth, means it turns one-third of the way around.

So one full turn of the small gear gives one-third of a turn of the big gear. The small gear turning fast makes the big gear turn slowly. The ratio between them is three to one.

But there is a trade. When the big gear turns slowly, it turns with three times as much force as the small gear was turning with. This is the deal that gears offer everywhere they appear, in cars and bikes and clocks and wind-up toys: **you can trade speed for force, or trade force for speed, but you cannot get more of both.**

If you have ever ridden a bicycle with multiple gears, you have used this trade-off without thinking about it. In a low gear, one push of the pedals barely moves you forward — but it doesn't take much effort to push, and you can get the bike moving even up a steep hill. In a high gear, each push of the pedals carries you a long way — but pushing is harder, and you can't really start in high gear from a standstill. Same legs, same bike, different deal.



Two meshed gears trade speed for force.

A box full of gears

A car has the same problem as a bicycle. Sometimes you need force (starting from a stop, climbing a hill, towing a trailer). Sometimes you need speed (cruising on the highway). One pair of gears can't do both.

So an old car has a box full of gears, with a way of choosing which pair is in use at any moment. The box is called the **transmission**, or the **gearbox**. (Both names are accurate. “Gearbox” tells you what it is. “Transmission” tells you what it does — transmits power from the engine.)

A typical 1960s car has four forward gears and one reverse gear:

- First gear: lots of force, very little speed. For starting from a stop.
- Second gear: less force, more speed. For getting up to about 20 miles per hour.
- Third gear: less force still, more speed. For cruising around town.
- Fourth gear: least force, most speed. For highway driving.

To switch between them, the driver moves a stick poking up out of the floor — the **gear shift**. Inside the transmission, the shift moves a small fork that slides one gear sideways into mesh with another. Different positions of the stick mesh different pairs of gears. The stick also has a position labeled R, for reverse, which engages an extra gear that flips the direction of rotation, so the car backs up.

That is the inside of the transmission. A box of metal gears, with sliders that decide which pair is doing the work at any given moment.

The clutch

There is still a problem. While the engine is running, the crankshaft is spinning all the time, at whatever speed the engine is running. If we hooked it up directly to the transmission, then any time the driver tried to switch gears, the spinning crankshaft would grind against the still gears in the new ratio, and we would shred the teeth off everything.

We need a way to temporarily disconnect the engine from the transmission while we shift gears. That is the **clutch**.

A clutch is, in its simplest form, two flat discs that can be either pressed together or pulled apart. One disc is bolted to the spinning crankshaft. The other disc is bolted to the input shaft of the transmission. When the discs are pressed together, friction holds them tight, and they spin as one — engine drives transmission. When the discs

are pulled apart, they aren't touching, and the engine spins on its own without doing anything to the transmission.

The driver controls the clutch with a pedal on the floor, called (unsurprisingly) the **clutch pedal**. Pedal up, discs pressed together, engine connected. Pedal down, discs pulled apart, engine disconnected.

So the procedure for changing gears looks like this. Push the clutch pedal down — engine disconnected. Move the gear shift to the new gear — different pair of gears slides into mesh. Let the clutch pedal back up — engine reconnected, now to the new ratio. All in a second or two. Repeat at every shift.

This is what your mom is doing in an old car when she does a little dance with her left foot every time the car changes speed. (Modern cars handle the whole clutch-and-shift dance automatically, with no clutch pedal at all. We will get to those in Part 2.)

Out to the wheels

The transmission has done its job. We have a shaft coming out the back of it, spinning at the right speed for the gear we are in.

In an old American car, the engine and transmission are at the front, but the wheels that get powered are at the back. So the spinning has to travel from front to rear. It does this through a long metal rod called the **driveshaft**, which runs along the underside of the car like a backbone.

The driveshaft is spinning. The two rear wheels need to spin. We are almost done. There is just one more clever piece.

When a car turns a corner, the two rear wheels travel different distances. The outside wheel goes around the outside of the curve (longer path), and the inside wheel goes around the inside (shorter path). If both wheels were forced to spin at the same speed, one of them would have to skid sideways through every turn.

So between the driveshaft and the two rear wheels sits a clever box of gears called the **differential**. It lets the two wheels turn at different speeds while still both being driven by the same driveshaft. The driveshaft turns the differential, the differential drives both rear axles, and the two axles can spin independently if the wheels are on different parts of a curve.

That is the last link in the chain. Crankshaft → clutch → transmission → driveshaft → differential → axles → wheels. The car moves.

TRY IT: BICYCLE GEAR RATIOS

You will need:

- A bicycle with multiple gears
- A way to flip it upside down (a bike stand, or just balance it on the seat and handlebars)

Turn the bike upside down so the wheels are off the ground and can spin freely. Look at the gears. The pedals turn a big toothed wheel called the chainring. The chain runs back to a smaller toothed wheel on the rear axle, which turns the rear wheel.

Pick the highest gear (largest chainring in front, smallest cog in back). Slowly turn the pedal one full turn with your hand and count how many times the rear wheel spins. Probably four or five times.

Now shift to the lowest gear (smallest chainring in front, largest cog in back). Turn the pedal one full turn again and count. Probably one and a half, maybe two times.

That difference is gear ratio. In the high gear, one pedal turn equals many wheel turns — great for going fast on a flat road, but very hard to start moving from a standstill. In the low gear, one pedal turn equals barely a wheel turn — great for getting up hills or starting from zero, but you can't go very fast.

Now flip the bike right-side up and ride it in each gear in a flat parking lot. Feel the difference. In the lowest gear, the pedals are easy but you barely move forward. In the highest gear, the pedals are harder to push but you can really cruise once you're up to speed.

A car works exactly the same way. The transmission is a multi-speed bike for the engine.

DID YOU KNOW? The differential — that clever device that lets the two rear wheels turn at different speeds — was not invented for cars. It is much older than that. Around 200 AD in China, engineers built a device called the south-pointing chariot. On top of the chariot was a wooden figure with an outstretched arm. Inside the chariot was a system of gears connected to the two wheels. As the chariot moved and turned, the gears compared the speeds of the two wheels, and used the difference to keep the figure's arm pointing south, no matter which way the chariot was rolling. It was a mechanical compass made entirely of gears. It worked by exactly the same math as the differential under a modern car. Some Chinese engineer in 200 AD figured out a piece of mechanical thinking that the rest of the world wouldn't put into a car for another seventeen hundred years.

Now we can go

So the wheels are spinning. The car is moving.

But it can't yet stop, and it can't yet steer. Two more things the car has to know how to do, neither of which has anything to do with the engine.

That is Chapter 5.

Stopping and Steering

The wheels are spinning. The car is moving forward.

Now we have a different problem. Maybe the most important problem on a moving car. The car has to go where the driver wants it to go, and stop when the driver wants it to stop. Otherwise it isn't a car. It is a runaway shopping cart with an engine.

This chapter is about the two systems that control where a moving car ends up: the brakes and the steering. The physics is completely different from anything in the last four chapters. No more explosions, no more sparks, no more gears. Just energy, friction, and a little geometry.

Stopping is energy

To stop the car, we have to deal with the energy it has.

A moving car has a kind of energy called **kinetic energy** — the energy of being in motion. Anything that is moving has it. A thrown baseball. A running dog. A speeding car. The faster something is moving, and the heavier it is, the more kinetic energy it has.

The exact amount is given by a simple formula. (One of the few equations in this book, but worth knowing.)

$$\text{energy} = \frac{1}{2} \times \text{mass} \times \text{speed}^2$$

The little ² at the end means “multiplied by itself.” Three squared is $3 \times 3 = 9$. Five squared is $5 \times 5 = 25$. Sixty squared is $60 \times 60 = 3,600$.

Here is what this formula tells us. If a car gets twice as heavy but moves at the same speed, its kinetic energy doubles. But if the car

stays the same weight and just moves twice as fast, its kinetic energy quadruples — because we squared the speed. Going from 30 miles per hour to 60 miles per hour means four times the kinetic energy, not twice. Going three times as fast means nine times the energy.

This is why high-speed crashes are so much worse than low-speed ones, and why a car going 60 needs four times the braking distance of a car going 30, not twice.

So here is the question. The car has all this energy of motion. To stop the car, you have to get rid of that energy. But energy doesn't just vanish. Where does it go?

It turns into heat

The answer: it turns into heat.

This is one of the deepest rules in physics. Energy can change from one form to another, but it cannot be created or destroyed. If a car has 100 units of kinetic energy at full speed and 0 units when it has stopped, that 100 units had to go somewhere.

Brakes work by deliberately creating friction between a part that spins with the wheel and a part that doesn't move. The friction slows the wheel down. It also makes the parts very, very hot. All the kinetic energy the car had a moment ago ends up as heat in the brake parts.

In a typical stop from highway speed, the brakes can heat up to several hundred degrees Fahrenheit — hot enough to fry an egg, several times over. In racing, they can hit far more than that. (See the sidebar.) That heat doesn't come from anywhere magic. It is the energy of motion the car had a moment ago, converted by friction into heat.

TRY IT: HANDS AS BRAKES

You will need: your hands.

Rub your palms together as fast and hard as you can for fifteen seconds. Then immediately put them against your cheeks or your forehead.

Your hands are warm. Maybe noticeably hot.

You did some work to rub them together — your muscles burned a little energy. That energy didn't disappear. It went into heat in your palms, through friction.

That is what brakes do. Except instead of two hands, it is a metal disc and two pads of friction material, with the kinetic energy of a 3,000-pound car moving at 60 miles per hour being shed as heat in just a few seconds. The brakes get extremely hot. They have to. There is nowhere else for the car's energy to go.

Two kinds of brakes

Two designs of brakes show up in old cars: drum brakes and disc brakes.

A **disc brake** has a flat metal disc, called the **rotor**, bolted to the wheel. The rotor spins with the wheel, all the time. Around the edge of the rotor sits a clamp called a **caliper** that doesn't move. Inside the caliper are two flat pads of friction material, one on each side of the rotor, with a small gap between them. When the driver presses the brake pedal, the caliper squeezes the two pads against the spinning rotor, pinching it from both sides. The rotor slows. The wheel slows with it. The pads get hot.

A **drum brake** is the opposite shape. Instead of a flat disc, there is a hollow drum bolted to the wheel — picture an upside-down cake pan. Inside the drum are two curved pads, called **brake shoes**. When the driver presses the pedal, the shoes get pushed outward, pressing against the inside surface of the spinning drum. Same idea, just inside-out.

By the late 1960s, most American cars had disc brakes on the front wheels and drum brakes on the back. (Disc brakes handle heat better, but cost more to make. The front brakes work harder than the rear, so cars got the better brakes where they mattered most.)

Either way, the principle is the same. A spinning part gets pinched or pushed against by a non-spinning part. Friction. Heat. The car slows down.

How the pedal makes it happen

Here is something you might not expect. The brake pedal is not connected to the brakes by a metal rod or a cable. The pedal pushes a piston in a small cylinder full of liquid. The liquid does the rest.

The system is called **hydraulic**. (From the Greek word for water, although brake fluid isn't water — it is a special oil that doesn't boil even when the brakes get hot.)

Here is why the system uses fluid. A liquid sealed inside a tube is almost impossible to compress. If you push one end of the tube with some force, the same force comes out the other end, instantly. Push harder, and exactly that much harder comes out the far end.

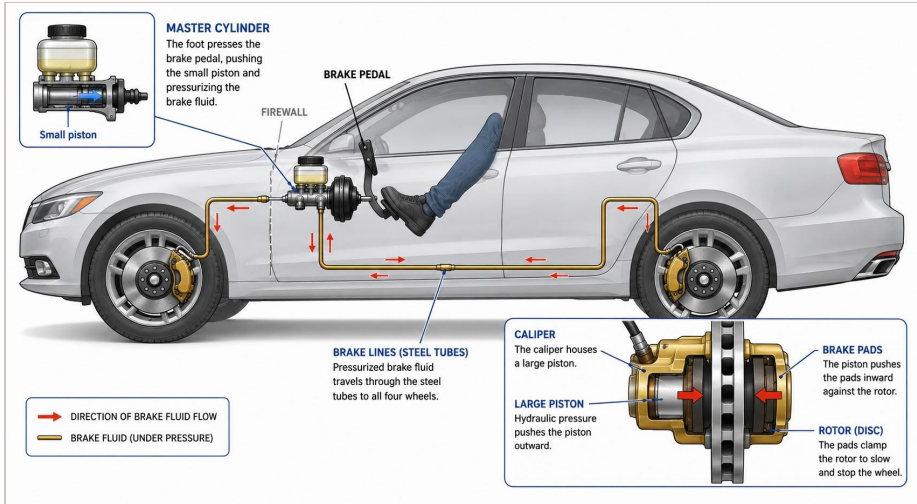
But there is a clever trick you can play with hydraulics that you cannot easily play with rods or cables. If the tube widens at one end, the force at the wide end gets multiplied. A small piston pushing into a small cylinder, connected through a tube to a larger cylinder with a bigger piston — push the small piston with five pounds of force, and the big piston pushes out with fifty.

(There is a catch. The big piston moves a much shorter distance than the small one. Just like gears trade speed for force, hydraulics trade distance for force. You don't get something for nothing. This rule keeps coming back.)

The brake system uses this trick. The driver presses a small piston with her foot. The pressure travels through small steel tubes to four

bigger pistons, one at each wheel. Those big pistons push the pads against the rotors. A modest push from a foot becomes a serious clamping force at each wheel.

When the driver lets the pedal back up, springs in each caliper push the pads back away from the rotor. The pressure drops. The fluid retreats through the tubes. The wheels spin free again.



How a hydraulic brake system works.

Steering

Stopping is about getting rid of energy. Steering is about geometry.

In a typical car, only the front wheels turn — the rear wheels just follow the path the front sets. The driver controls the front wheels through a chain of mechanical parts.

The steering wheel in the driver's hands connects to a long shaft called the **steering column**, which runs down from the wheel to the bottom of the car. At the bottom, a box of gears (or, in newer cars, a clever toothed bar called **rack and pinion**) converts rotation of the steering wheel into side-to-side motion. From there, a pair of metal

arms called **tie rods** push or pull each front wheel into the angle the steering wheel is asking for.

Turn the steering wheel left, the tie rods pull the front wheels left, and the car follows. Right, same thing in reverse.

There is a geometric subtlety. When a car turns a corner, both front wheels pivot — but they don't pivot the same amount.

Picture a car making a tight left turn. The left front wheel is on the inside of the curve, taking a tight path. The right front wheel is on the outside, taking a wider path. For the wheels to roll cleanly through the turn without scrubbing sideways, the inside wheel has to pivot at a sharper angle than the outside wheel. (If they both pivoted by the same amount, one of them would be forced to skid sideways through the turn — same problem the differential solved for the rear wheels in the last chapter.)

The steering linkage handles this automatically. The arms that push each wheel are arranged at angles so that turning the steering wheel tilts the inside wheel more than the outside wheel, by exactly the right amount. The driver doesn't think about it. She turns the wheel, and the right amount of tilt arrives at each front wheel on its own.

Putting your back into it

In an old car — really old, before about the 1960s — there was no help with steering. The driver had to provide all the muscle to push the front wheels into a turn, especially when the car wasn't moving. Parallel parking an old car without help is a workout. You are literally pushing two heavy wheels sideways through gravel or asphalt.

Modern cars use **power steering**, which uses either hydraulic pressure or an electric motor to assist the driver. In a power-steering car, even a kid could turn the wheel of a parked car with one hand. Without it, even an adult has to lean into it.

Many 1960s American cars came with power steering as an option. By the 1980s it was standard on most cars. By the 2010s, almost all of it had switched from hydraulic to electric. We will meet electric power steering in Part 2.

DID YOU KNOW? In Formula 1 racing, where cars routinely brake from 200 miles per hour down to 50 in a few seconds, the brake discs glow orange-red after every hard braking — visibly hot at night. Each disc can briefly hit 1,800°F, hotter than lava. Race teams use brake discs made of carbon-ceramic, a special material that can survive temperatures that would melt steel. Even with that material, the discs slowly wear away through a season of racing — the kinetic energy of every braking corner has to go somewhere, and a tiny amount of disc material goes with it as smoke and dust each time the brakes are used.

End of Part 1

That is the entire mechanical car. Engine making power through controlled explosions; carburetor mixing fuel and air; ignition system making the spark; transmission and drivetrain getting the power to the wheels; brakes and steering controlling where the car ends up.

Put all of that together, and you have a 1960s car. The kind your grandparents drove. Every part of it is a piece of metal pushing or pulling another piece of metal, in ways you could see and touch if you took the car apart. No software. No computers. Just physics, levers, gears, and gasoline.

Then, sometime in the 1970s and 1980s, things started to change. The first small computers appeared under the hood, doing one job each. The mechanical parts didn't go away — they are still there, even in the newest cars on the road today — but they got managers. Sensors watching them. Software adjusting them in real time. And new sys-

tems started showing up that the mechanical parts could not do alone: brakes that don't lock up in a panic stop, fuel mixtures that change with the weather, computers that watch the road and pull the steering wheel for you if you're about to drift out of your lane.

That is Part 2.

PART TWO

How Cars Got Smart

The Brain

Everything in Part 1 was a piece of metal pushing or pulling another piece of metal.

The carburetor mixed air and fuel because air rushing through a narrow throat created low pressure, which sucked fuel up through a hole. There was no decision in there. It was just shaped metal and physics.

The ignition system fired sparks at the right time because a rotating shaft inside the engine pushed two pieces of metal apart at exactly the right moment. Again — no decision. Just shaped metal and timing.

Mechanical solutions like these were clever, but each one could only do one thing. A carburetor can mix fuel and air well, but it cannot change the mixture if the air outside gets thinner. A distributor can fire sparks in order, but it cannot decide that this particular cylinder needs the spark a tiny bit later than the next one.

To make a car that can change its behavior — that can adjust to what is happening — you need something that can think.

That something is a computer.

Why a car needs a brain

Two things happened in the 1970s that pushed cars toward needing computers.

First, the air. American cities in the 1960s had terrible smog — yellow-brown haze, hard to see across, hard to breathe. A lot of it came from cars. Burning gasoline produces several different chemicals as

exhaust, some of which are bad for the air. A perfectly mixed engine burns much more cleanly than a poorly mixed one. But a mechanical carburetor can't always hit the perfect mix. It does its best, and its best isn't quite good enough.

Second, the gas. In 1973, gasoline got expensive, fast. People wanted cars that could go farther on a gallon. Like emissions, fuel economy depends on burning the fuel-air mixture exactly right, every time, in every condition — and again, mechanical systems weren't precise enough.

The fix was to replace the carburetor with **fuel injection** — small sprayers that squirt a metered amount of gasoline directly into the intake at exactly the right moment — and to control those sprayers with a computer that could measure conditions and adjust on the fly. The mechanical distributor got similar treatment: replaced by electronic spark control, where the computer decides exactly when each spark should fire.

By the early 1980s, every new American car had a small computer under the hood. Today, a typical car has dozens of them, each handling a different job.

The big one — the one that runs the engine — is called the **engine control unit**, or ECU. (Some cars call it an engine control module, or ECM. Same thing.) This chapter is about how the ECU works.

Senses

A computer cannot make decisions about something it cannot see. So the first thing the ECU needs is **sensors** — devices that measure something physical and turn it into an electrical signal the computer can read.

A modern engine has somewhere between fifteen and thirty sensors, depending on the car. Each one watches one thing.

The throttle position sensor watches how far down the gas pedal is pressed and tells the ECU how much power the driver is asking for. The mass airflow sensor sits in the intake and measures how many grams of air are flowing into the engine each second. The engine speed sensor reads the crankshaft and reports the current RPM. The coolant temperature sensor watches how hot the engine has gotten, so the ECU knows whether the engine has warmed up yet.

The cleverest one is probably the oxygen sensor. It sits in the exhaust pipe and measures how much oxygen is left in the burned gases on their way out. By measuring leftover oxygen, the ECU can tell whether the last batch of fuel was perfectly mixed, too rich, or too lean. It is like tasting your soup after cooking, to see whether the seasoning came out right.

There are more — knock sensors listening for engine pinging, camshaft position sensors, intake pressure sensors, and others. Each one feeds the ECU a different stream of numbers, dozens of times per second. Together, these sensors are the car's senses. Without them the ECU is blind.

The brain itself

In the middle of all those sensor wires sits the ECU. From the outside, it looks like nothing — a small metal box about the size of a hardback book, usually bolted to the inside of a fender or near the dashboard. Inside is a circuit board, with a few microchips and some other electronic parts.

The most important chip on the board is the **microprocessor** — a small piece of silicon that can do simple arithmetic millions of times per second. (For comparison: your phone's processor is about a hundred times faster than the one in an engine ECU. The ECU's chip is much slower than your phone's, but it doesn't need to play games or

stream videos. It just needs to make engine decisions, fast and reliably.)

Inside the microprocessor's memory is the program — a long set of instructions the engineers wrote — and a big table of numbers called a **map**. The map is essentially a giant lookup table. For every combination of engine conditions, the map says: “given this airflow, this RPM, this temperature, fire the spark this many degrees before the piston reaches the top, and inject this many milliseconds' worth of fuel.”

When the engine is running, the ECU's program does roughly this, in a loop, dozens of times per second:

1. **Read all the sensors.** Get current numbers for airflow, RPM, temperature, oxygen, and so on.

2. **Look up the right values in the map.** Match the current conditions to the prescribed response.

3. **Make small corrections.** If the oxygen sensor says the last batch was a touch too rich, lean it out a touch; if too lean, richen it.

4. **Send commands to the actuators.** Tell the fuel injectors how long to spray and the ignition coils when to fire.

Then it does it all again. And again. And again. About a hundred times per second, every second the engine is running.

Muscles

Sensors are useless unless the computer can act on what they tell it. So just as the car has senses, it has **actuators** — devices that take commands from the ECU and do something physical in the engine.

The two most important are the fuel injectors and the ignition coils. **Fuel injectors** are small electric valves, one per cylinder, that spray a precise amount of gasoline into the cylinder when the ECU commands them to. The ECU controls each injector independently, telling it exactly how long to stay open. A few milliseconds longer means a richer shot of fuel; a few milliseconds shorter means leaner.

The ignition coils are the same coils we met in Chapter 3 — but in a modern car, instead of one big coil and a mechanical distributor, each cylinder usually has its own little coil. The ECU fires each one electronically, at exactly the moment it wants that cylinder to spark.

There are other actuators too — idle valves, variable timing solenoids, throttle servos in cars where the gas pedal is electric. But fuel injectors and ignition coils are the two big ones. Together, they let the ECU control the two most important things about combustion: how much fuel goes in, and exactly when it ignites.

A tour around the loop

Here is the whole thing in motion. Your mom is driving up a hill. The car needs more power than it would on flat ground, because gravity is fighting it. Watch what the ECU does, all in less than a second:

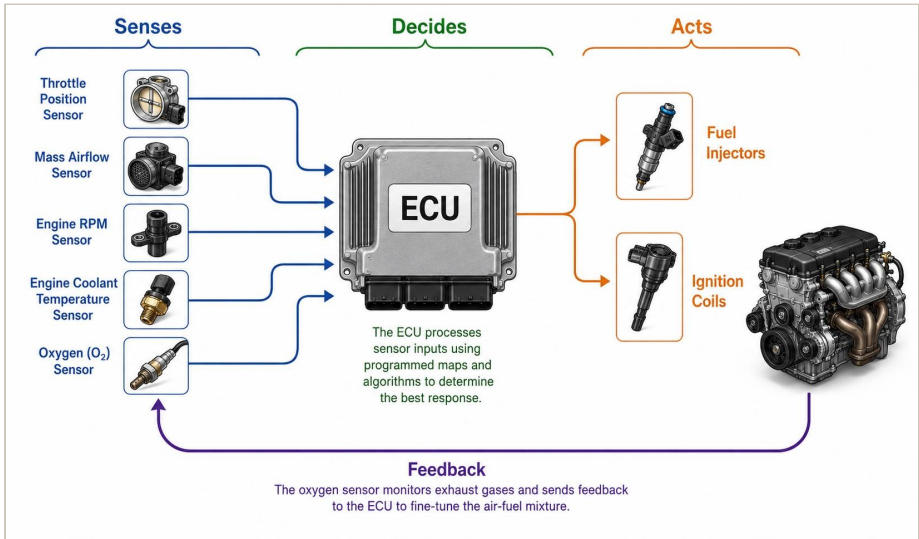
Your mom presses the gas pedal a little farther down. The throttle position sensor sees the change and reports it to the ECU. The throttle plate in the intake opens wider, more air starts flowing into the engine, and the mass airflow sensor reports the new airflow number.

The ECU looks up the new conditions in its map: more air, same RPM, what fuel and timing should I use? It finds the answer. It sends slightly longer pulses to the fuel injectors so they stay open a few milliseconds longer per shot, putting more fuel in to match the extra air. It nudges the ignition timing a tiny bit to get the most power out of the bigger fuel-air charge. The bigger charge ignites, the engine makes more power, the car climbs the hill.

Meanwhile, the oxygen sensor in the exhaust is reading the burned gas on its way out, telling the ECU whether the mixture turned out a touch too rich or a touch too lean. The ECU nudges the next batch a hair the other way.

This loop runs continuously, hundreds of times per minute, all without the driver knowing anything is happening other than that the

car is responding to her foot. The ECU has just done what a 1960s carburetor and distributor could not — adjusted, on the fly, in real time, based on what was actually happening, to get the best possible burn.



Sense, decide, act: the ECU loop.

TRY IT: THE TOILET TANK LOOP

This is going to sound weird. Trust me.

You will need:

- A toilet
- An adult's permission to lift the tank lid

Lift the lid off the back of a toilet tank and set it carefully on the floor. Inside, you will see water, a chain, a flapper valve at the bottom, and a tall plastic pole or cylinder with a float on it. Now flush. Watch what happens.

When the handle goes down, the chain lifts the flapper, water rushes out into the bowl, and the level in the tank drops. As the level

drops, the float drops with it. Once the float gets low enough, a valve opens at the top of the supply pipe and fresh water starts pouring in. The water level rises again. When the float rises high enough, the valve closes. The water stops. The tank is full.

What you just watched is a complete sense-decide-act loop, the same kind the ECU runs. The float is the sensor — it measures the water level by sitting on top of it. The mechanism that opens and closes the valve is the decider — it decides “valve open below this level, valve closed above this level.” The valve itself is the actuator — it does the work of letting water in or stopping it.

A toilet tank is not smart. It has no microprocessor, no software, not a single transistor. But it has senses, decisions, and actions, all linked in a loop. That is exactly what the ECU is doing inside your car, just faster, with electronics, and with stakes higher than overflowing porcelain.

DID YOU KNOW? Long before computers, engineers solved control problems with clever mechanical linkages. The most famous example came from James Watt, the Scottish engineer who improved the steam engine in the late 1700s. Steam engines had a problem: if they ran too fast, they could shake themselves apart. Watt's solution, in 1788, was a device called a flyball governor. It was two metal balls on hinged arms, attached to a vertical shaft that spun along with the engine. As the engine sped up, the balls spun faster, flew outward (the same way you fly outward on a fast carnival swing-ride), and pulled levers that closed the steam valve. As the engine slowed, the balls fell back inward and the valve opened more. A purely mechanical, self-correcting control loop, working entirely on its own. Sensor: the flyball arms. Decider: the linkage geometry. Actuator: the steam valve. The same architecture as a modern ECU, made entirely of metal, two hundred years before microchips existed.

The car can adapt

The car has a brain now. Sensors at one end to feel what is happening, actuators at the other end to do something about it, and the ECU in the middle reading and deciding hundreds of times per second. The mechanical heart is still there — same pistons, same crankshaft, same combustion — but a layer of decision-making has been wrapped around it. The car can adapt.

What can it do, now that it can think?

A lot. The next chapter is about safety systems — anti-lock brakes, traction control, stability control, airbags. Every one of them is the same sense-decide-act loop, pointed outward at the world instead of inward at the engine, and used to help prevent crashes.

That is Chapter 7.

Crash Prevention

We left Chapter 6 with a question. What can the car do, now that it can think?

This chapter answers part of that question.

The most useful thing the computer does in a modern car is watch for trouble. It watches the wheels. It watches the road. It watches how the car is moving through space. And when something starts to go wrong, it reacts in milliseconds — much, much faster than any human driver could.

The result is that drivers today get into fewer crashes, and when they do crash, they walk away more often. Modern cars are remarkably hard to roll, hard to spin out, hard to skid past their braking limit. None of that is because the laws of physics changed. It is because there are small computers wedged into the corners of the car, watching every wheel hundreds of times per second, ready to do something faster than you can blink.

This chapter is about those computers and what they do.

How tires grip the road

To understand what these safety systems are doing, you have to understand one fact about tires.

A tire grips the road by friction — the same friction that made your hands warm when you rubbed them in Chapter 5. The rubber tread of the tire pushes against the bumps and roughness of the pavement, and grips. Up to a point, this grip is rock solid. The tire rolls along,

and the road pushes the tire forward (when you accelerate), backward (when you brake), or sideways (when you turn).

But there is a limit.

Push too hard — brake too hard, accelerate too hard, turn too sharply — and the tire crosses a line. Instead of gripping the road, it slips. The rubber starts sliding across the pavement instead of rolling along it. And here is the key fact: a sliding tire grips much less than a rolling one.

You can feel this with your sneakers. On a smooth floor, you can walk normally — your shoe grips, you go. But if you try to stop suddenly while running on the same floor, your shoe might skid out from under you. Once a shoe is sliding, it has way less grip. Same physics.

Every safety system in this chapter exists for one reason: **to keep the tires on the right side of that line.** To keep them rolling, not sliding. Below the limit, not over it.

Watching the wheels

Three of the most important safety systems in a modern car all rely on this same principle. Each one watches for a different way the tires might cross the slipping line.

Anti-lock brakes (ABS) catches the case where you brake hard and a wheel is about to lock up. When a wheel locks, it stops rolling and starts sliding — and as we just said, sliding gives way less grip than rolling. The car actually takes longer to stop with locked wheels than with rolling-but-barely wheels. Before computers, drivers were taught to “pump the brakes” — press and release rapidly — to keep the wheels just barely rolling. ABS does this automatically, dozens of times per second, on each wheel independently. If a wheel slows down faster than the car is, ABS releases the brake pressure on that wheel for a tiny moment, lets the wheel start rolling again, and then

re-applies. The driver feels this as a rapid pulsing in the brake pedal. That pulsing is ABS doing its job.

Traction control watches the opposite case. When you press the gas hard, the engine pumps power into the drive wheels. If the road is icy or wet or covered in loose gravel, the wheel can spin in place — sliding around its axle without gripping the road. That is bad for two reasons. The car doesn't go forward, and a spinning tire has nearly no grip if you also need to steer. Traction control uses the same wheel sensors as ABS. If a drive wheel suddenly spins faster than the others, the computer cuts engine power for a moment, or applies a brief brake to that wheel, until it starts gripping again.

Stability control watches the trickiest case. Sometimes the whole car starts to slide sideways — the back end fishtails out in a turn, or the front end pushes wide. To catch this, stability control adds another sensor: a small device that measures how fast the car is rotating around its own vertical axis (engineers call this **yaw**). If the car is rotating more or less than the steering input says it should, the computer applies the brake to one wheel only — usually for less than a second — to twist the car back into alignment. Pulling on just the right front wheel, for example, drags the right side of the car backward, which yaws the car right and can correct a left-sliding skid.

All three systems use the same wheel-speed sensors — small magnetic devices that count the teeth on a ring spinning with each wheel, hundreds of times per second. All three feed into the same computer. All three are different branches of one big watchful system, each catching one of the three ways a tire can lose grip. Each one is a sense-decide-act loop, exactly like the ECU from the last chapter — just pointed at the wheels instead of at the engine.

When prevention fails

Sometimes the crash happens anyway. A driver doesn't see a stopped car ahead in time. A pedestrian steps off a curb. A truck runs a red light.

A different system takes over. Airbags don't try to prevent the crash. They make the crash less awful for the people inside.

Embedded in the front of the car are sensors that measure deceleration — how fast the car is losing speed. Most of the time the deceleration is small (your mom hits the brakes at a stoplight). But in a real crash, a car can go from 30 miles per hour to zero in about a tenth of a second. That is an enormous deceleration, and it doesn't look anything like normal braking. The sensor recognizes it and signals the airbag computer.

The airbag computer makes one decision: deploy.

From that moment, things happen very fast. An electrical pulse fires a small igniter inside the airbag module. The igniter sets off a chemical reaction (more in the sidebar). The chemical reaction produces a huge amount of nitrogen gas in milliseconds. The gas inflates a folded fabric bag, which bursts out of the steering wheel or the dashboard.

About fifty milliseconds — one-twentieth of a second — pass between the sensor recognizing the crash and the bag being fully inflated. By the time the driver's body, still moving forward at the original speed, would have hit the steering wheel, there is a fabric pillow there instead.

The airbag does not stay inflated for long. Within another tenth of a second, vents in the bag let the gas out, and the bag absorbs the impact as it deflates. The airbag isn't there to push you back. It is there to absorb the energy of you flying forward. Once it has done that, it gets out of the way.

Watching the road

The newest safety systems do not just watch the car. They watch the world around it.

A camera mounted near the rearview mirror reads the lane lines on the highway. If the car drifts toward a line without the turn signal on, the camera notices, and the computer either nudges the steering wheel back or shakes the seat to wake the driver up. This is **lane-keep assist**.

A radar in the front bumper measures the distance to the car ahead, and how fast that distance is shrinking. If the closing speed is dangerous and the driver hasn't started braking, the computer slams on the brakes itself. This is **automatic emergency braking**.

Sensors in the rear corners watch for cars in the next lane and light up a small icon in the side mirror when one is there. **Blind-spot monitoring**.

All of these are sense-decide-act loops, exactly like the ECU and the wheel-watching systems. The sensors are different — cameras and radar instead of magnetic wheel-speed sensors — but the architecture is the same. Sense, decide, act. Faster than a human can react.

TRY IT: CATCH THE RULER

You will need:

- A 12-inch ruler
- A friend or parent

Stand up and hold your dominant hand out in front of you, with your thumb and index finger open about two inches apart, ready to pinch closed.

Have your friend hold the ruler vertically so the bottom end of it is just above your fingers, with the 0-inch mark at the bottom and the 12-inch mark at the top. Don't touch it yet.

Without any warning, your friend drops the ruler. The instant you see it start to fall, snap your fingers shut to catch it.

Look at the number where you grabbed it. That tells you how far the ruler fell before you reacted. If you caught it at the 4-inch mark, your reaction took about 0.14 seconds. At the 6-inch mark, about 0.18 seconds. At the 8-inch mark, about 0.20 seconds. If the ruler hit the floor, you were too slow — about half a second. (Try again.)

Most people fall in the 0.15-to-0.25 second range. That is how long it takes a human to see something happen and start to do something about it.

Now compare. ABS makes its decisions in less than 0.01 seconds — about twenty times faster than you. By the time you noticed your wheels were locking up, ABS would have already pulsed the brakes twenty or thirty times, on each wheel independently. The airbag controller decides to deploy in about 0.015 seconds. Stability control measures the car's rotation a hundred times per second.

These computers aren't smart in the way you are. They cannot decide what to have for dinner, or argue with their parents, or read a book. But for one specific job — reacting faster than a human can react — they are wildly better than you, or your mom, or any driver who has ever lived.

DID YOU KNOW? Inside an airbag, before it fires, is a small canister of solid chemical, usually sodium azide. (Pronounced AY-zide.) An electrical pulse from the airbag computer ignites the chemical. The reaction is nearly instant. In a few thousandths of a second, the solid chemical decomposes into sodium metal and pure nitrogen gas — a lot of nitrogen gas, fast. Nitrogen makes up about 78% of the air we already breathe, so it is harmless. But producing that much of it that fast is, technically, a small explosion. The fabric airbag absorbs the explosion and turns it into a controlled inflation, and the bag is fully open in about one-twentieth of a second. Most of the dust people see when an airbag deploys is talc and cornstarch — engineers add a little of each to keep the folded bag from sticking to itself before it fires. It is harmless. But it does look like the bag is smoking, which freaks people out.

A car full of computers

So the modern car is full of small computers, all of them watching for different kinds of trouble. The ECU watches the engine. The braking computer watches the wheels for skidding. The stability computer watches the chassis for unwanted spinning. The crash computer watches for sudden deceleration. The lane camera watches the road lines. The radar watches the car ahead.

A typical modern car has, depending on how you count, somewhere between fifty and a hundred small computers built into it. Each one does its own job. None of them are very smart on their own.

But they need to talk to each other — to share what they see, to coordinate their actions.

How they do that is the next chapter.

On the Same Wire

We left Chapter 7 with a problem. A modern car has fifty to a hundred small computers in it. Each one does its own job. None of them are very smart on its own.

But many of the things a car does well require several computers to cooperate. The stability control system from the last chapter — the one that catches the car if it starts to slide sideways — needs to know how much the driver has turned the steering wheel (from the steering computer), how fast each wheel is spinning (from the brake computer), how hard the engine is pulling (from the ECU), and how fast the car is rotating around its own axis (from the yaw sensor). It uses all four of those numbers, many times per second, to figure out whether the car is going where it should be going.

That information lives in four different places in the car. The stability computer needs to gather it all.

To do that, the computers need to talk to each other.

This chapter is about how they do.

Wires everywhere

The first idea you might have is straightforward. Just connect every computer to every other computer with a wire. The brake computer wires directly to the stability computer. The ECU wires directly to the dashboard. The lane camera wires directly to the steering. And so on.

This works for two computers. It works for three, with three connections. For four, you need six connections. For ten, forty-five. For a

hundred — the number in a typical modern car — you need 4,950 connections.

That is a lot of wires.

It is also a lot of weight (copper is heavy), a lot of cost, a lot of plugs, and a lot of places where something can come loose. Modern cars don't do this.

One shared line

What modern cars do instead is run two thin wires through the entire car, and connect every computer to those same two wires. Every computer talks on the same shared line. The shared line is called a **bus**. In a car, the most common one is called the **CAN bus**, short for Controller Area Network. It was invented in the 1980s by a German engineering company called Bosch.

Here is how it works. Every message sent on the CAN bus has two parts: an ID number (which says what the message is about — “wheel speeds,” “engine RPM,” “steering angle,” and so on) and the data itself (the actual values).

When a computer wants to send a message, it puts the message on the wire. Every other computer on the bus receives it — every single one. The computers that don't care about that message just ignore it. The computers that do care read the message and act on it.

It is like the public address system at a school. The PA broadcasts an announcement to every classroom in the building. Every teacher and every student hears it. Most of them ignore it. But the kids it is actually for — “Will the third grade please report to the gym?” — pay attention. The PA doesn't have to address each student personally. It sends one message, and the listeners filter.

The CAN bus is the PA system of the car. One channel, every computer listening, each one filtering for what it cares about.

Taking turns

There is one obvious problem. If every computer can speak on the same wire, what happens when two of them try to speak at the same time?

They have to take turns.

The CAN bus has a clever rule for deciding who goes first. Every message has a priority number (which is, conveniently, just the same as its ID). When two computers try to talk at the same time, the one with the higher-priority message wins, and the lower-priority one waits a fraction of a millisecond and tries again.

Important messages — engine timing, brake commands, anything safety-critical — get high priority. Less urgent messages — door lock states, dashboard light statuses — get lower priority. The bus organizes itself, with no central controller and no traffic cop, just a rule that makes the right thing happen automatically.

Each message is short. About eight bytes of data — enough for a handful of numbers. The CAN bus runs at half a million bits per second, so dozens of messages can fit into a single millisecond. Plenty of bandwidth to keep all the computers in sync.

Six computers, half a second

Here is what this looks like in real time. Your mom is driving down a wet highway, and the car suddenly starts to slide sideways through a curve. Watch what happens — all in less than a second:

1. The yaw sensor detects that the car is rotating faster than the steering wheel input would suggest. It puts a message on the bus: “yaw rate higher than expected.”
2. The stability computer, which has been listening for that message, picks it up. It already knows, from messages it received earlier in the same second, the steering angle (from the steering computer), the

wheel speeds (from the brake module), and the engine torque (from the ECU).

3. The stability computer does some quick math and decides the right rear wheel needs a brief, firm brake to pull the car back into line.

4. The stability computer sends a message on the bus: “brake module: please apply firm pressure to the right rear caliper for 200 milliseconds.”

5. The brake module, which has been listening for messages addressed to it, picks up that command. It applies the brake.

6. Meanwhile, the dashboard module — which is also listening to the bus — sees the “stability control activated” message and lights up a small icon to let the driver know the car is helping.

7. A fraction of a second later, the yaw sensor reports that the spin has stopped. The stability computer tells the brake module to release. The dashboard turns off the icon.

Total elapsed time: maybe 200 milliseconds — about as fast as you can blink. Six different computers, communicating across one pair of wires, working together to keep the car going where it is supposed to go. None of them in charge of the others. Each one doing its own job, listening to and producing messages on the same shared line.

That is the modern car.

One bus is not enough anymore

For most of the 1990s and 2000s, a single CAN bus could handle every computer in a car. But cars kept getting more computers, and the original CAN bus is too slow for some modern needs. (Half a million bits per second is plenty for engine and brake messages. It is not nearly enough for a high-resolution camera streaming video at 30 frames per second.)

So modern cars have several different buses, each tuned for what it carries. The original CAN bus still handles engine, brake, and stability

messages. A slower, cheaper bus called LIN handles low-priority things like the buttons in your door panel and the rearview mirror motors. A faster bus called FlexRay handles ultra-time-critical safety. And for cameras and self-driving sensors, modern cars use Ethernet — the same kind of network that connects computers in an office.

The car has gone from one shared wire to something more like a small office network on wheels.

TRY IT: THE CONVERSATION GAME

You will need:

- Two or more family members
- A dinner table or other place where everyone is together

The rules are simple. Each person takes turns saying a short message. Every message must start with who it is for. Some examples:

- “Mom: please pass the salt.”
- “Dad: did you remember to call the dentist?”
- “Everyone: dinner is ready.”
- “Whoever has the dog: she's at the back door.”

Now play for a few minutes. Notice what happens. Everyone hears every message — you are all sharing the same room and the same air. But only the addressed person is expected to respond. If a message starts with “Mom,” dad doesn't answer. If it starts with “Everyone,” everyone is supposed to react.

Notice also: when two people try to talk at the same time, one of them stops and lets the other go first, then takes a turn after. People do this naturally, without thinking.

You have just acted out, with humans instead of microchips, the basic protocol of the CAN bus inside your mom's car. Shared channel. Addressed messages. Everyone hears everything. Receivers filter what is relevant. Talkers take turns. Higher-priority things go first.

The car does this on two wires, with computers, hundreds of times per second. You did it with your family for five minutes at dinner. Same idea. Same architecture. Different speeds.

DID YOU KNOW? Underneath the dashboard of every car sold in the United States since 1996, there is a small port called OBD-II. It looks like a 16-pin plug shaped a bit like a trapezoid, usually tucked under the steering column or near the driver's left knee. You probably haven't noticed it. Mechanics use it constantly. Through that one port, a special tool called a scanner can listen to everything happening on the car's CAN bus — every wheel speed, every fault code, every sensor value, in real time. If your mom takes the car to the shop because the check-engine light is on, the very first thing the mechanic does is plug a scanner into the OBD-II port. The car tells the scanner what is wrong, in plain text codes. (P0301, for example, means “cylinder 1 misfire detected.”) Before OBD-II, mechanics had to guess at problems. After OBD-II, they have a direct line into the car's nervous system. One small plug under the dash, one window into the entire car.

End of Part 2

That is Part 2.

The modern car is a hybrid of old and new. The mechanical chain from Part 1 is all still there — pistons firing, the crankshaft turning, the transmission ratcheting through gears, the brakes converting energy to heat, tie rods pivoting the front wheels. None of it has gone

away. Take the hood off a modern car and you'll see the same kinds of moving metal that filled the hood of a 1960s car.

But all of it is now wrapped in software. Sensors watching every part. Computers reading the sensors and deciding things hundreds of times per second. A network of buses tying them all together. The old physics is still doing the work. The new electronics are managing the work.

Then, in the early 2000s, an old idea — much older than most people realize — came back. People stopped putting gas in some cars at all. Some cars now have no engine, no transmission, no gears, no exhaust pipe. They have a giant battery and an electric motor, and not much else.

That is Part 3.

PART THREE

How an Electric Car Works

The Big Battery

For the first eight chapters of this book, every car we have looked at has run on gasoline. That gasoline got mixed with air in either a carburetor (Part 1) or a fuel injector (Part 2), squeezed by a piston, lit by a spark, and burned. The hot expanding gas pushed the piston down, the piston turned the crankshaft, the crankshaft turned the wheels.

For an electric car, you can throw all of that out.

There is no gasoline. No cylinders. No pistons. No spark plugs. No transmission with gears. No long driveshaft. No exhaust pipe. No fuel pump.

What an electric car has, instead, is a giant battery (under the floor) and an electric motor (between the wheels), connected by wires.

That is it. Practically the whole car.

This part of the book is about how an electric car works. We start with the battery, because the battery is the heart of an electric car the same way the engine is the heart of a gasoline car. It is the biggest, heaviest, most expensive thing in the car. And it stores all the energy the car will ever use until somebody plugs it in again.

What a battery is

A battery stores energy as chemistry, and gives it back as electricity.

Inside a battery are two pieces of different material — usually two different metals, or chemicals containing metals — kept apart by something that conducts ions but not electrons. (Don't worry too much about the words “ion” and “electron” yet. The point is that the

separator lets parts of atoms move through it, but not the electricity itself.) When you connect the two pieces by an outside wire, a chemical reaction happens between them, and electricity flows through the wire while it is happening.

Run the reaction one way, and electricity comes out — that is discharging the battery, the way you do every time you use it. Run the reaction the other way (by pushing electricity in from outside), and the battery stores up energy again — that is charging.

Every battery, from a watch battery to a Tesla pack, works on this same principle. Two different materials, separated by something that lets ions through, hooked up by an outside wire that carries the actual electric current.

The materials and design have been refined over more than two hundred years. The materials in a modern car battery (or phone, or laptop) are lithium and a few cleverly chosen compounds. The basic idea is the same as the very first battery, invented in 1800. (See the sidebar.)

Why the battery is huge

A car uses a lot of energy to drive.

To go 250 miles, an electric car typically needs to store somewhere between 50 and 100 kilowatt-hours of energy. That number doesn't mean much without comparison, so:

50 kilowatt-hours is roughly the amount of electricity an average American home uses in two days. It is the energy in about a gallon and a half of gasoline. It is enough work to lift a small car twenty miles into the air.

All of that has to fit inside the battery, in lithium-ion chemistry. There is no way around the laws of physics here. Chemistry stores a certain amount of energy per kilogram of material, and lithium-ion happens to store about 250 watt-hours per kilogram. To hold 50 kilo-

watt-hours, you need at least 200 kilograms — about 440 pounds — of battery cells. Real EV battery packs typically weigh 800 to 1,200 pounds total once you include the housing, cooling system, wiring, and management computer.

That is why electric cars are heavy. The battery is heavy. There is no shortcut.

A gasoline tank, for comparison, is light — gasoline holds about 12,000 watt-hours of energy per kilogram, almost fifty times more than a lithium-ion battery. That is why a single gallon of gasoline goes such a long way, and why electric cars couldn't compete with gas cars on range until batteries got really, really good.

Cells, modules, and the pack

The actual battery in an EV is not one big block. It is made of thousands of little batteries, called cells, packed together like sardines.

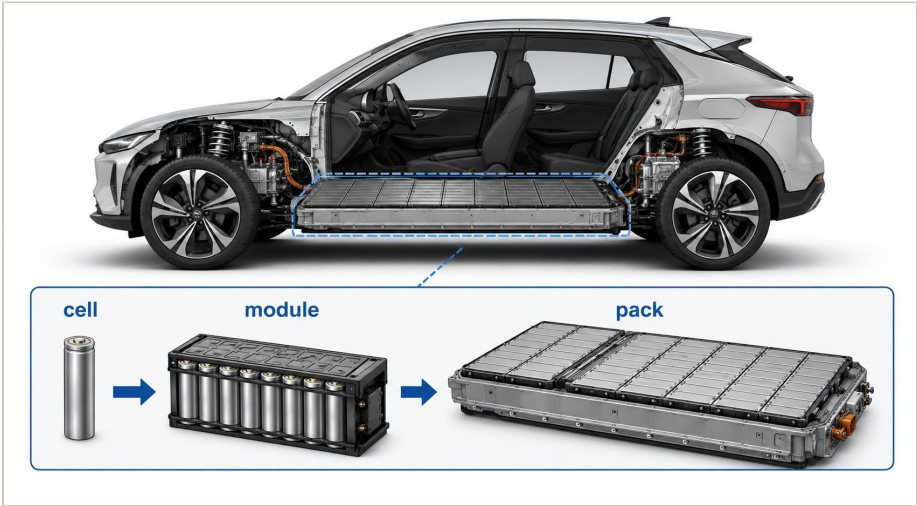
A single lithium-ion **cell** looks more or less like an AA battery, except in different shapes. Some cars use cylindrical cells (like AA's, but bigger and fatter — about the size of a roll of nickels). Some use rectangular pouch cells (like a thick deck of cards). Some use prismatic cells (like a small juice box). They all do the same thing: they hold about three or four volts of electrical pressure across their two terminals, and somewhere between two and ten amp-hours of capacity. (Roughly: enough energy to power a small light bulb for a few hours.)

To get from a cell to a car battery, engineers do two things. Stack cells end to end, and their voltages add together. Three cells of three volts each, in a stack, give nine volts. This is called wiring them **in series**. EV battery packs typically have many cells stacked in series to get a total voltage between 300 and 800 volts. Yes — eight hundred volts. That is much higher than the outlets in your house (which are 120 volts), and high enough to be very dangerous if anyone tried to

mess with it. EV battery packs are sealed and shielded for exactly this reason.

Side by side, cells share the load and add their capacities. Two cells of three amp-hours each, side by side, give six amp-hours. This is called wiring them **in parallel**. EVs use parallel wiring to get total capacity high enough to drive hundreds of miles between charges.

Combine series and parallel, and you get a battery with both high voltage and high capacity. A typical EV battery is around 400 volts, with somewhere between 4,000 and 8,000 individual cells, all wired in a careful series-parallel arrangement under the floor of the car. The whole assembly — cells plus the metal housing plus the cooling system plus the wiring plus the management computer that watches every cell — is called the **battery pack**, or just the pack.



From cell to module to pack.

Putting energy back in

To use the battery, you discharge it — pull energy out, drive somewhere, repeat. Eventually it runs low. Then you have to put energy back in. There are two main ways to charge an EV.

Slow charging uses regular wall power. You plug the car into your house, the way you plug in a phone. Inside the car is a small device called the **onboard charger**, which takes the alternating-current (AC) power from the wall, converts it to direct-current (DC) power that the battery can store, and feeds it slowly into the pack. Charging from empty might take 8 to 20 hours, depending on the car and the outlet.

Fast charging uses a special station that bypasses the onboard charger. The station itself converts the power to DC and feeds it directly into the pack at much higher rates — 50, 150, or even 350 kilowatts. At a fast charger, an EV can go from 20% to 80% charge in about 30 minutes.

Why doesn't every charger fast-charge? Two reasons. Fast chargers cost a lot to build — they need very beefy power converters. And charging too fast, too often, slowly damages the battery. Slow charging at home is gentler on the chemistry, and most EV owners do most of their charging that way, overnight in a garage. Fast charging is for road trips.

What batteries don't like

Lithium-ion batteries are remarkable, but they have weaknesses.

They don't like cold. When the temperature drops below freezing, the chemistry inside the cells slows down, and the battery delivers less power and accepts less charge. Driving an EV in a Minnesota winter typically loses 20 to 30 percent of the car's range, until the battery warms up. Many EVs include heaters that warm the pack while you drive, partly to fix this problem.

They don't like extreme heat. Above about 110 degrees Fahrenheit, the chemistry starts to degrade faster, and the battery ages more quickly. EVs include cooling systems — often using the same kind of liquid-cooling loops you find in a high-end gaming PC — to keep the pack at a comfortable temperature.

They don't last forever. Every time you charge and discharge a lithium-ion cell, a tiny amount of capacity is lost. After about a thousand charge cycles, a cell typically holds about 80 percent of what it started with. After several thousand, much less. EV batteries are designed to last about ten years of normal driving, after which the car still works, but its range is shorter than when it was new.

These limits are real, but they are not deal-breakers. They are tradeoffs. The pack is heavy. The car charges more slowly than a gas car refuels. The range drops in winter. In exchange: no engine, no fuel system, no exhaust, almost nothing to maintain, and a remarkably smooth ride.

TRY IT: THE LEMON BATTERY

You will need:

- A fresh lemon (or four, if you don't have a multimeter)
- A galvanized nail (zinc-coated, from any hardware store)
- A copper penny made before 1982 (anything older is mostly copper) or a piece of bare copper wire
- A multimeter, if your family has one — or some bare copper wire if you don't

Roll the lemon on a hard surface for a minute, pressing down with your hand. This breaks up the juice sacs inside. Now stick the nail into one end of the lemon, halfway in. Stick the penny (or copper wire)

into the other end, halfway in, far enough from the nail that they don't touch inside the lemon.

You have just built a battery.

If you have a multimeter, set it to read DC volts and touch one probe to the nail and the other to the penny. The meter will read somewhere between 0.5 and 0.9 volts. That is real, measurable electricity, made by the chemistry of zinc and copper reacting through the lemon juice.

If you don't have a multimeter, the easiest way to detect the electricity is to use four lemons. Build four identical lemon batteries, then connect them in a chain — penny of lemon 1 to nail of lemon 2, penny of lemon 2 to nail of lemon 3, and so on, using small bits of bare copper wire to make the connections. The four-lemon chain produces about 3.5 volts. Touch the unconnected nail at one end of the chain and the unconnected penny at the other end to your tongue at the same time. You will feel a faint, distinctive tingle. The current is far too small to hurt you. You are tasting electricity made out of chemistry.

The lemon battery works on the same principle as an EV battery: two different metals separated by something that lets ions move between them, hooked up by an outside path through which current flows. Scale it up — better materials, ten thousand cells in series and parallel — and you have a Tesla pack.

DID YOU KNOW? The battery in your phone uses the same basic chemistry as the battery in an electric car. Both are lithium-ion batteries — they store and release energy by moving lithium ions back and forth between two electrodes. The phone battery is about the size and shape of a credit card and stores enough energy to run your phone for a day. A car battery is the size and shape of a small mattress and stores enough energy to drive a car several hundred miles. Same chemistry, scaled up by a factor of about a thousand. The whole idea of a battery goes back further. The very first one was invented in 1800 by an Italian scientist named Alessandro Volta. His version was a stack of zinc and copper discs separated by saltwater-soaked cardboard. The stack made about 1.5 volts of electrical pressure — exactly the same as a modern AA battery in a TV remote. Volta's stack worked. Every battery since, including the one in your mom's phone and the one under the floor of every Tesla, is a refinement of his idea.

Energy in a box

So we have a battery. Hundreds of pounds of carefully arranged lithium-ion cells, wired in series and parallel to make a few hundred volts and many thousands of amp-hours, sitting in a sealed box under the floor of the car.

That battery holds energy. It does not yet do anything with it.

To turn the energy into a moving car, we need a motor.

That is the next chapter.

The Motor

Last chapter, we built the battery. Hundreds of pounds of lithium-ion cells, sealed under the floor of the car, holding all the energy the car will use until somebody plugs it in again.

That energy is just sitting there. To make the car go, we need to turn the stored chemistry into motion.

The device that does that is the motor. And here is something surprising about the motor.

It is much, much simpler than a gasoline engine.

A gasoline engine, for all its cleverness, has hundreds of moving parts. Pistons. Connecting rods. Valves. Camshafts. A crankshaft. A timing belt. Spark plugs firing thousands of times a minute. Oil sloshing through every joint. An electric motor has, depending on how you count, somewhere between one and a few moving parts. There is a thing that spins, and a thing that doesn't. That is mostly it.

This chapter is about how that thing spins.

Electricity in a coil makes a magnet

We have to take a short detour back to physics before getting to the motor.

In Chapter 3, we mentioned in passing that running electricity through a coil of wire makes the coil act like a magnet. We used this fact for the ignition coil — current through the inner coil made a magnetic field, and a sudden collapse of that field induced a giant voltage in the outer coil.

It is worth pausing on the first half of that sentence. **Electricity in a coil makes a magnet.** This is one of the most useful facts in physics. Wrap any old piece of wire around any old metal core (a nail, a finger, even nothing at all), hook the ends to a battery, and you have built a magnet. Disconnect the battery, the magnetism vanishes. Reverse the battery's connections, and the magnet's north pole becomes its south pole and vice versa.

This is what makes electric motors possible. If you can build a magnet that you can turn on, off, and flip, you can use it to make things move just by sending electricity in.

Pushing with magnets

If you have ever played with two magnets, you know the basic deal. North faces north, they push apart. North faces south, they snap together. The strength of the push depends on how close they are and how strong each magnet is. With strong magnets, the push is enough to throw them across a tabletop.

Imagine one magnet that you can spin freely on a pivot, with another magnet held still nearby. The free magnet will swing around to align itself — opposite poles facing each other — and then settle there. Once they line up, it stops moving. The free magnet has reached its happy place.

This is not yet a motor. To make it a motor, we need a way to keep the free magnet from settling — a way to make it keep wanting to rotate.

How a motor actually spins

Here is the trick. You make the spinning magnet not a permanent magnet, but an electromagnet — a coil of wire that becomes a magnet only when current flows through it. Then you do this:

1. Send current through the coil. The coil becomes a magnet, with one end north and the other end south.

2. The coil swings around toward alignment with the fixed magnet, the way the free magnet did in the last section.

3. Just before the coil settles into alignment, you reverse the current.

4. The coil's north and south ends instantly swap places. Now the coil is pointed the wrong way, and the fixed magnet pushes and pulls it through another half-rotation.

5. Just before it settles again, you reverse the current again. And so on.

The coil keeps swinging around, never quite reaching its happy place because the moment it gets close, you flip the rules on it.

That is a motor. A coil of wire, on a pivot, near a magnet, with the current cleverly flipping every half rotation.

The thing that does the flipping in an old-fashioned motor is called a **commutator**. It is a clever pair of metal contacts on the coil's spinning shaft. As the shaft turns, those contacts mechanically reverse the direction of current at exactly the right moment. No software, no controller — just shaped metal doing the right thing as the shaft rotates past.

Modern EV motors don't use commutators. Instead, a small computer called the **motor controller** reads the position of the spinning part many times per second and feeds current through the coils in exactly the right pattern at the right moment. This is more efficient, has fewer moving parts that can wear out, and lets the controller play other tricks too — like running the motor faster or slower with great precision, or turning it from a motor into a generator (more on that next chapter).

Stator and rotor

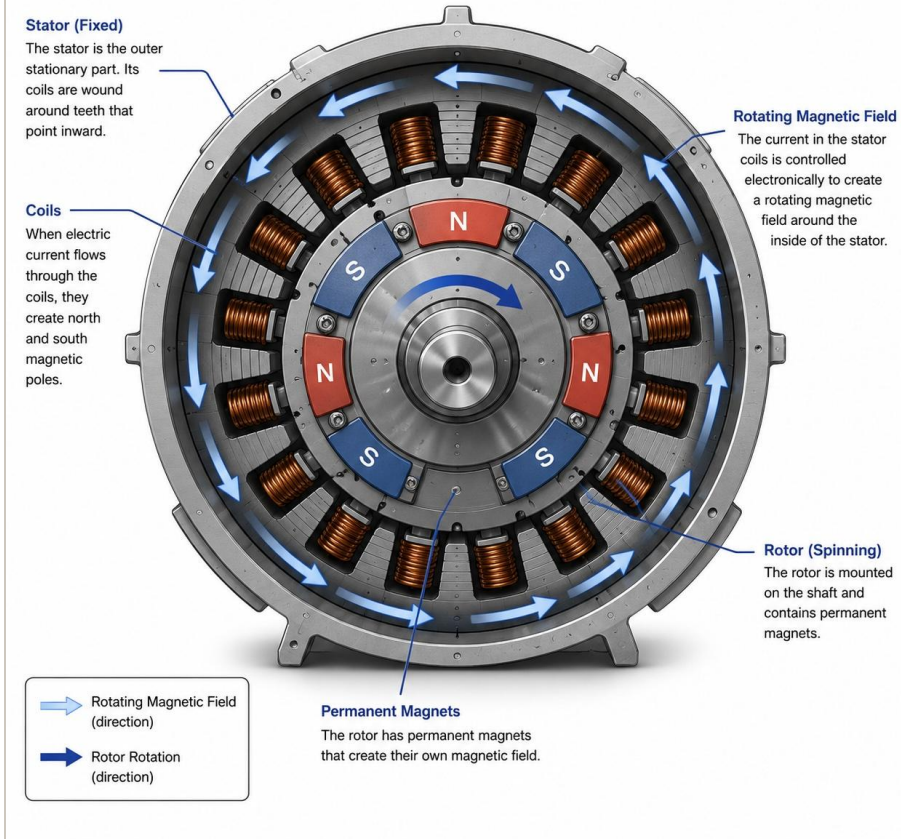
Inside an EV motor are two main pieces. The **stator** is the part that doesn't move. It is a hollow cylinder lined on the inside with multiple coils of wire (usually three sets, called the three **phases**). The motor controller sends current through these coils in a careful, rotating pattern.

The **rotor** is the part that spins. It sits inside the hollow cylinder of the stator, on a shaft. In the most common type of EV motor, the rotor has its own permanent magnets bolted onto it.

When the controller sends current through the stator coils in the right pattern, the magnetic field inside the stator rotates — the field is moving around the inside of the cylinder without anything physically moving. The permanent magnets on the rotor try to follow that rotating field, the way a compass needle would follow a moving magnet. The rotor spins. The rotating field stays just ahead of it. The rotor chases it forever, or at least until the controller cuts the current.

The shaft of the rotor connects to the wheels through one fixed gear ratio. Spin the rotor, and the wheels turn.

EV Motor – Cross Section



Cross-section of an EV motor.

Why no transmission

The fact that the motor's speed and force are entirely under the control of the controller has a remarkable consequence.

An electric motor produces full force from zero RPM.

A gasoline engine cannot do this. As we saw in Chapter 4, a gas engine has to be turning at at least a thousand RPM to make any meaningful power. Below that, it stalls. That is why a gasoline car has

a clutch and a multi-speed transmission — to bridge the gap between an engine that wants to spin fast and wheels that are stopped.

An electric motor doesn't have that gap. As soon as you press the pedal, the controller starts sending current through the stator. The rotor immediately wants to spin, even from a complete standstill, and it pushes against the wheels with full force right away. As you press the pedal further, the controller sends more current, and the rotor spins faster. The pedal directly controls how hard the rotor is pulling — and the pull is there from the very first moment you press.

This is why an electric car doesn't need a transmission. There is no clutch, no gear shift, no neutral, no first or second or third gear. Just one fixed gear ratio between the motor and the wheels, and the controller takes care of everything else.

It is also why electric cars feel so quick to accelerate from a stop. A gas car at a green light has to spin up its engine, engage the clutch, and ease into first gear before the wheels get any real push. An electric car at a green light gives you full force the instant the pedal moves. No spinning up. No gear-engaging. Just the rotor going from still to fast in milliseconds.

TRY IT: BUILD A SIMPLE MOTOR

This one takes a little more setup than most of the others, and might want a parent's help. The materials are cheap and easy to find online or at a hardware store.

You will need:

- About 3 feet of enameled copper wire (also called “magnet wire” — it has a thin colored coating along its whole length)
- 1 fresh AA battery
- 2 large paper clips
- A small permanent magnet (a refrigerator magnet works; a small “rare earth” neodymium magnet works much better)
- A piece of fine sandpaper
- Tape or a rubber band

Wrap the copper wire into a tight coil around something cylindrical (the AA battery itself works) — make 15 to 20 loops. Slide the coil off, then wrap each loose end of the wire around the coil twice to hold the bundle together. Leave each end sticking straight out, like the axle of a wheel, about an inch on either side.

Now the trick. The enamel coating on the wire is what insulates it — current can only flow where the coating is removed. With the sandpaper, gently sand the coating off the **BOTTOM HALF** of each wire end. Leave the **TOP HALF** coated. Do the same on the other wire end, and make sure both wires are sanded on the same side (both undersides). This is the trickiest step. Get a parent to check your work before moving on.

Bend each paper clip so the top has a small loop you can rest a wire in. Tape one paper clip to each end of the AA battery, with the loops sticking up. Place the magnet on top of the battery, between the paper clips, with one of its faces pointing up.

Rest the two wire ends of your coil into the paper clip loops, so the coil is suspended above the magnet. Give the coil a small starting nudge with your finger.

It should keep spinning on its own.

Here is what is happening. When the bare side of each wire is touching the paper clip, current flows from the battery through the coil. The coil becomes a magnet. The permanent magnet on top of the battery pushes the coil — and the coil flips around. As it rotates so the coated side of the wire is touching the paper clip, current stops flowing (the coating insulates), but the coil keeps spinning from its own momentum. When the bare side comes back around, current flows again, and the magnet pushes again. The half-on, half-off pattern of your sanded wire is acting as a tiny commutator, switching the current at exactly the right moments to keep the coil spinning.

Sometimes it takes a few tries — the coil has to be balanced, the wire ends have to make good contact, the magnet needs to be strong enough. If your motor doesn't spin, try a stronger magnet, more loops in the coil, or a fresh battery.

You just built a working electric motor. The one in a Tesla is bigger, fancier, and has a computer controlling it instead of sanded wire — but the principle is identical.

DID YOU KNOW? Electric cars are not new. The first one was built in the 1830s, around the same time as the first electric motors. By the late 1800s, electric cars and gasoline cars were both on the streets, competing for customers. In 1900, in New York City, electric cars actually outsold gasoline cars. What sank the early electric cars was the battery. A lead-acid battery from 1900 stored only a fraction of the energy of a modern lithium-ion pack — about a tenth — and it was just as heavy. Range was 30 to 40 miles between long, slow charges. A gasoline car could go 200 miles and refuel in five minutes. The choice was not close. Electric cars faded almost completely by the 1920s. The motor side of an electric car was already perfectly fine in 1900. The car your great-great-grandparents might have ridden in had an electric motor that, in principle, would still work today. It was the battery that needed another century of work. Once lithium-ion came along, electric cars came back.

Energy moving

So now we have the full architecture of an electric car. Battery in the floor, holding chemical energy. Motor at the wheels, turning electrical energy into rotation. A motor controller deciding how much current to send through the stator coils, based on how far the driver has pressed the pedal. Wires connecting all of it.

That is how an electric car drives.

There is one more clever thing — something a gas car can't really do. When the driver lifts off the pedal or starts to brake, the motor doesn't just stop pushing. It can briefly run backwards, pulling energy back out of the moving car and sending it into the battery. The same device that drives the wheels can act as a generator, putting energy back instead of taking it out.

That is the next chapter.

Energy Back

Six chapters ago, in Chapter 5, we asked a question. A 3,000-pound car going 60 miles per hour has an enormous amount of kinetic energy — one-half mass times speed squared, you might recall. To stop the car, you have to get rid of that energy. Where does it go?

In a gasoline car, the answer was: it gets turned into heat in the brake pads, and then that heat dissipates into the air. The energy is gone. The car burned gasoline to gain that energy in the first place. Then, every time it stopped, it deliberately threw the energy away.

There is a Try It from that chapter where you rubbed your palms together hard for fifteen seconds and felt them get warm. That is what brakes do. The chapter ended with a line: there is nowhere else for the car's energy to go.

It turns out, in an electric car, there is.

You can put the energy back in the battery.

The system that does this is called **regenerative braking**. It is one of the cleverest tricks in any modern car, and it follows from a single piece of physics we have already met.

A motor and a generator are the same device

Back in Chapter 3, we used this rule of physics:

A collapsing magnetic field through a coil of wire creates a voltage in that coil.

At the time, we used it to explain how the ignition coil makes 20,000 volts to fire a spark plug. But the rule is more general than that.

Any time a magnetic field changes through a coil — getting stronger, getting weaker, moving past, doesn't matter — a voltage appears in the wire.

The motor we built in the last chapter uses the OTHER direction of this rule. We send a current through a coil, the coil becomes a magnet, and that magnet pushes against another magnet to make rotation.

What if you ran the motor backwards?

Imagine you have an electric motor — coils inside a stator, magnets on a rotor — but instead of feeding current to the coils, you take the shaft of the rotor and spin it yourself. The permanent magnets on the rotor sweep past the coils on the stator. From the coils' point of view, magnetic fields are constantly changing — getting stronger as a magnet approaches, weaker as it leaves, then the next magnet does the same.

By Chapter 3's rule, that means voltages appear in the coils.

If you connect those coils to a battery (or any other electrical thing), current flows out of them. The motor, when it is being spun, is making electricity.

This is not a different device. It is **the same device**. A motor and a generator are the same device. The difference is which direction you run it: feed current in, get rotation out; feed rotation in, get current out. It is one of the most elegant symmetries in physics, and it is the entire foundation of regenerative braking.

How it works when you brake

Picture an electric car driving down the road at 30 miles per hour. The motor controller is feeding current to the stator coils, the rotor is spinning, and through one fixed gear ratio, the wheels are spinning along with it.

Your mom takes her foot off the pedal.

The motor controller stops feeding current to the coils. But the car doesn't stop instantly — it has all that kinetic energy, the wheels are still rolling, the rotor is still spinning. From the wheels, the motion is be-

ing fed back through the gear ratio to the rotor, just as it always was.

So now the rotor is spinning, the stator coils are no longer being fed current, and Chapter 3's rule is still in effect. The rotor's magnets are sweeping past the coils, magnetic fields are changing, and voltages are appearing in the coils.

The motor controller has a choice. It can leave the coils disconnected, in which case the voltages just sit there and the car coasts (slowing only from air drag and rolling friction). Or it can connect the coils to the battery.

If it connects them, the induced voltages drive current INTO the battery. The battery starts to charge. And here is the critical part: pulling current out of the coils makes the rotor harder to spin. The same physics that gives you a generator effect also creates a drag on the rotation.

The car slows down. The battery gains a little energy. The kinetic energy that would have become wasted heat in the brake pads is captured back.

That is regenerative braking. The car's kinetic energy travels through the wheels, through the gear ratio, through the rotor, through the stator coils, through the controller, and into the battery. The reverse of the trip it took on the way out.

One-pedal driving

In a gasoline car, the driver uses two pedals to control speed. The right one (gas) makes the car go faster. The left one (brake) makes it go slower. Lifting off the gas just lets the car coast — slowing very gradually from air drag, but not really braking.

In a modern EV, the accelerator pedal does both jobs.

Press it down: the motor pulls power from the battery. The harder you press, the more current flows, the more force the motor makes, the harder the car accelerates.

Lift it most of the way off: the motor controller switches the system into generator mode. The further you lift (the closer to fully off), the more current flows back into the battery, and the more drag the motor produces on the wheels. The car slows down — sometimes hard enough to come to a near-stop without anyone ever touching the actual brake pedal.

This is called **one-pedal driving**, and many EV drivers swear by it. After a few weeks of it, drivers stop using the brake pedal except for emergencies and the last mile per hour or two of stopping. The accelerator does both adding and subtracting energy, with no transition.

It also makes the car much smoother to drive. There is no jump between gas-pedal physics and brake-pedal physics. It is all one continuous control. Press a little harder, accelerate a little harder. Lift a little, decelerate a little. How far the pedal is from fully pressed directly determines how energy is moving between the wheels and the battery — in either direction.

What regen can't do

Regenerative braking is not magic. It has real limits.

It can't capture 100% of the energy. Some of the kinetic energy is always lost — as heat in the motor windings, as heat in the wires, as friction in the bearings. Real EVs typically recover somewhere between 50% and 70% of the energy of any given stop. The rest still becomes heat, just less of it than in a gas car.

It can't always run. If the battery is already fully charged, there is nowhere to put the captured energy, and regen has to be reduced or shut off. The car falls back on its friction brakes. The same is true if the

battery is too cold to accept fast charging. EV computers watch this constantly and switch smoothly between regen and friction braking based on what the battery can do at that moment.

It can't bring the car to a full stop on its own. As the rotor slows, the rate of magnetic field change drops, and the induced voltage drops with it. Below a few miles per hour, regen has almost no slowing effect. The friction brakes — the same disc-and-pad ones from Chapter 5 — have to handle the last bit of every stop.

It can't help in an emergency. If a driver slams on the pedal because something is in the road, the car needs maximum stopping force right now, and that means friction brakes do almost all the work. Every EV still has a complete friction brake system at all four wheels for exactly this reason.

What regen does very well is the everyday case: gentle slowing, stopping at red lights, coasting down hills, easing through traffic. In those situations it captures most of the energy that would otherwise be lost. Across a full day of normal driving, regen typically extends an EV's range by ten to twenty percent — with the bigger gains coming on the slow, stop-and-go streets where braking happens often, and almost no gain on the highway, where the car spends most of its time cruising and rarely brakes at all.

TRY IT: SPIN A MOTOR BACKWARDS

This Try It works best if you still have the motor you built in Chapter 10. (If you don't, any small DC motor — the kind inside many electric toys — will work.) You will also need a multimeter.

Set the multimeter to read DC volts on its lowest range (often labeled “200 mV” or similar). Touch one probe to each lead of the motor. The reading should be at or near zero. Now spin the motor's shaft as fast as you can with your fingers — pinch it between thumb and forefinger and twist.

Watch the multimeter. The number jumps. It might only be a few millivolts (thousandths of a volt), but it is real, and it is coming from your spinning. The faster you spin, the higher the voltage.

You have just turned the motor into a generator, by hand.

Now connect the motor's leads to a small battery — like the AA battery you used in the original build — and watch what happens. The motor spins on its own. (Same device, opposite direction.)

If you happen to have a small low-voltage LED, you can try the more dramatic version: connect the LED across the motor's leads (LEDs only light when current flows in one direction, so if it doesn't light, swap the leads). Spin the shaft fast. The LED flickers. You are powering an LED with the motion of your fingers, through a generator that costs about \$2 to build.

Scale this up — bigger motor, stronger magnets, the wheels of a car instead of your fingers — and you have an EV's regenerative braking system.

DID YOU KNOW? Subway trains have been doing regenerative braking for over a hundred years. Electric subway trains run on power supplied by an overhead wire or a third rail. When a train brakes, its motors switch to generator mode (the same trick we just covered), and the recovered electricity flows BACK into the rail. Another train somewhere on the network — usually one that is currently accelerating — can pull that electricity right out of the rail and use it. Twenty to forty percent of a modern subway's electricity comes from the regenerative braking of other trains on the same line. If you ride the New York City subway, the London Underground, or any major metro system, you are riding inside a giant regenerative braking network. Every time the train you are on slows down, you might be helping power another train, miles away, that is just leaving its own station.

That is the electric car

Battery in the floor. Motor at each driven axle. Controller deciding, second by second, when to push current out (to drive) and when to pull it back (to brake). A few hundred pounds of lithium ions, a few coils of copper wire, a few permanent magnets, and the math of changing magnetic fields. From those ingredients, a car.

This is where cars are, in 2026.

Where they go next is the last chapter.

By the Time You Drive

You are about ten years old. In five or six years you will probably learn to drive. Maybe seven, depending on where you live.

The car you first drive will not be the same as the car your mom drives today. Cars keep changing, and 2032 or 2033 is a long time away in car years.

This chapter is about a few things that are already changing, and a few things that might.

I am going to be careful here. Predicting the future of cars has a long history of being completely wrong. In 1900, smart people said there would never be more than a few thousand cars in the United States. In 1985, smart people said the year 2000 would have flying cars. They didn't. So instead of guessing, I will tell you what is actually happening in 2026, and let you watch how it plays out.

Bigger and faster batteries

The most certain change is in the battery. Lithium-ion has been getting steadily better for thirty years — about 7% more energy per kilogram each year, on average. That doesn't sound like much, but it adds up. The battery in a 2025 EV holds roughly twice the energy of one from 2010, in the same space.

Two specific things will keep that going. The first is **solid-state batteries**. The lithium-ion batteries in today's cars (and your phone) use a liquid called electrolyte to let lithium ions move between the two electrodes. Solid-state batteries replace that liquid with a solid

material — usually a special glass or ceramic. They are smaller for the same energy, charge faster, and don't catch fire as easily as liquid ones. Companies have been promising them for over a decade. As of 2026, only a few prototype cars actually have them. By the time you drive, most new EVs probably will.

The second is **faster charging**. The fastest charging stations in 2026 can take an EV from 20% to 80% in about 18 minutes. (That is a lot quicker than the 8 to 20 hours your home charger needs.) Within another five or ten years, charging an EV at a highway stop will probably be about as fast as a long bathroom break — maybe ten minutes for a full road-trip charge. Once that happens, one of the biggest practical objections to EVs will go away.

Cars that drive themselves (sort of)

The big question for the next decade is how much of the driving the car will do for you.

The honest answer in 2026 is: more than ten years ago, less than the headlines suggest.

A typical new car can already do many things on its own. It can keep itself in its lane on a highway. It can match the speed of the car in front and brake to a stop in traffic. It can park itself in a tight spot. It can warn the driver about cars in the blind spot, pedestrians stepping into the road, and obstacles up ahead. (Most of these were the safety systems from Chapter 7.)

What it generally cannot do, in 2026, is drive entirely without you watching. Every car sold for personal use in 2026 expects a human driver to be paying attention at all times, ready to take over. The systems are very good helpers. They are not yet pilots.

Self-driving taxis, on the other hand, do exist. In a few cities — Phoenix, San Francisco, parts of Los Angeles — companies operate cars without drivers, picking people up the way Uber drivers do.

These cars have spinning sensors on their roofs and a dozen cameras around their bodies. They drive better than most people in some ways (they never get distracted) and worse in others (they sometimes get confused by construction zones, snow, or people doing weird things in the road). They run only in cities they have been carefully mapped for. They are not yet everywhere.

Whether self-driving cars become normal in your lifetime depends on hard problems that haven't all been solved. A computer driving in 99% of situations is doable. A computer driving in 100% of situations — every weird corner case, every unusual weather condition, every time someone does something the software didn't predict — is much, much harder. The difference between 99 and 100 is enormous.

By the time you drive, your car will probably be able to do more for you than your mom's car can today. Whether you will ever own a car that drives itself completely is genuinely uncertain.

The car that powers your house

Here is something you might not have heard about. Some EVs can run in reverse — but not in the way you might think. They can pull power out of their giant battery and feed it back into the house.

Remember from Chapter 9 that an EV battery holds 50 to 100 kilowatt-hours of energy. A typical American home uses about 25 kWh of electricity per day. Which means a fully charged EV holds enough energy to power a house for two to four days.

Some EVs already support this. The Ford F-150 Lightning and a few others can be plugged into a special outlet that lets the car power your house during a blackout. Charge the car overnight when electricity is cheap, drive around during the day, and at night it powers your fridge and lights, especially if the grid goes down.

The next step is more interesting. Imagine if every EV on a street could, all at once, push a little power back onto the grid when the grid

needed it — say, on a hot summer afternoon when air conditioners are pulling more electricity than the power plants can supply. A few hundred EVs charging at home can match the output of a small power plant. Some companies and cities are starting to experiment with this. The technical name for it is **vehicle-to-grid**, or V2G. By the time you drive, your car might not just be a car. It might also be part of a giant battery for your city.

Maybe fewer cars

The other big change is harder to see, but it might matter most. People in many cities are using cars less.

E-bikes — bicycles with small electric motors that help you pedal — are exploding in popularity. They go about as fast as a slow car (around 20 miles per hour), don't need a driver's license, don't need parking, and cost a tenth what a car costs. In Amsterdam, Copenhagen, and increasingly in places like Brooklyn and Portland, more and more people just don't use a car much in the city. They take an e-bike or a regular bike, ride to a train station, take the train downtown, and ride to the office.

Ride-sharing apps — Uber, Lyft, the self-driving taxis we just talked about — let people get rides without owning anything. Some young people are already choosing not to get a driver's license at all, because they don't see the point. Cities are improving public transit — better trains, better buses, more reliable schedules.

Will you own a car when you grow up? Possibly. Probably, if you live in most of the United States, where towns are spread far apart. But maybe not, especially if you live in a city. The thing called a “car” in 2040 might still be a personal vehicle that lives in a garage, or it might be a service that picks you up when you ask, or it might be an e-bike that you ride to a train. Cars are not going to disappear, but the question of how big a part of your life they are is genuinely changing.

TRY IT: COUNT EVS

On your next car ride with your mom, take a notebook. Mark down every car you can identify as electric or hybrid for the whole trip.

Some clues that a car is electric or hybrid:

- A Tesla logo (a stylized “T”) on the front or back. All Teslas are electric.
- A small plug symbol on the back, often near the brand name.
- The words EV, Electric, Hybrid, or Plug-in printed on the back.
- Brand names like Polestar, Rivian, or Lucid — these companies make only electric cars.
- Some specific models that are always electric or hybrid: Toyota Prius, Honda Insight, Chevy Bolt, Nissan Leaf, Ford Mustang Mach-E.

Over a 15- or 20-minute drive, count how many you see, and how many gas-only cars you see. The proportion will tell you what kind of neighborhood you are in. Some places are well over 30% EVs already. Some are still under 5%.

Now do this once a year for the next few years. The number will keep going up. By the time you drive, EVs probably will not be unusual at all. They will just be cars.

DID YOU KNOW? Around 1900, a journalist in New York City wrote that the streets would soon be impossibly clogged with horse manure. Horses were everywhere. They pulled milk wagons, buses, police carts, family carriages. Each horse produced about 25 pounds of waste per day. New York's streets were so filthy that walking them in nice shoes was a daily problem. Then, between roughly 1900 and 1920, almost all the horses disappeared from American cities. The car had arrived. The whole change took about twenty years. To people in 1900, gasoline cars were a strange and untrustworthy novelty — loud, unreliable, expensive. By 1920, they were everywhere, and horses had been pushed almost completely out of the cities. People who had said in 1905 that their grandkids would still ride horses to school turned out to be wrong. The transition you are watching now — from gasoline cars to electric ones — is similar in some ways and different in others. It is happening more slowly (probably 30 to 40 years rather than 20). It is happening unevenly (much faster in China and Norway than in the United States). And it might never finish completely; some gas cars will probably stick around for a long time. But the basic shape — a new technology gradually replacing an old one because the new one is better in enough ways — is one that has happened many times before.

The box is open

At the start of this book, I said we were going to open the box. The car as a sealed mystery, just a thing that moves with you sitting inside it.

The box is open now.

The pistons are firing. The mixture is being mixed. The spark is bouncing across the gap inside the cylinder. The wheels are turning through gears and a driveshaft and a differential. The brakes are converting energy into heat. The computers are watching every wheel hundreds of times per second. Or — in the electric version of all this

— the battery is sending current to the motor, the motor is pushing the wheels, and when the car slows down, the energy is going back the other way.

Whatever the cars look like by the time you drive, the same physics applies. Energy stored somewhere gets converted into motion. Tires grip the road through friction up to a limit. Brakes do something with kinetic energy, whether they make heat or store it back into a battery. Steering uses geometry to point the wheels. Computers watch and decide and act.

Pistons and crankshafts and carburetors might all eventually become history, the way horses became history. But a car will still be a metal box on four wheels, with seats and windows and a steering wheel, taking people from one place to another. The basic experience of being a person inside a car has been the same for over a hundred years. It will probably be the same in 2050.

You started this book as a passenger. You still are. But now you know what is happening around you when you ride.

You don't drive yet. That's fine. You don't need to drive to understand a car, the same way you don't need to fly a plane to understand how a wing works. You just need to pay attention.

Now you know what to pay attention to.